Generative models for Images & Videos

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OxML 2025

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Generative Modeling

Simply put:

- Learn the distribution of the data p(x)

Why?

- Learn useful representations
- Sample from it

Generative Modeling

Training

Learns p(x)

$\mathbf{x}$

$\mathbf{z}$

Constructed by hiding information from x

Model

$\hat{\mathbf{x}}$

Loss

Target

Generative vs. Discriminative

Image credit - Wikimedia

Generative Modeling

Training

Learns p(x)

$\mathbf{x}$

$\mathbf{z}$

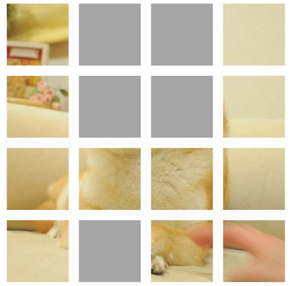
Constructed by hiding information from x

Model

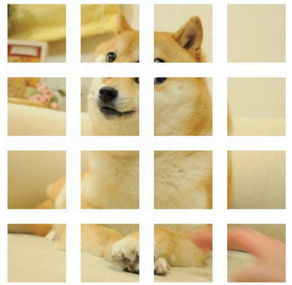
$\hat{\mathbf{x}}$

Loss

Target



Masked Patches



Target

Generative Modeling

Training

Learns p(x)

$\mathbf{x}$

$\mathbf{z}$

Model

$\hat{\mathbf{x}}$

Loss

Constructed by hiding information from x

Target

Inference

Model

Encode a “x”

Model

Sample from p(x)

This Talk

Mainly about diffusion/Flow Matching models

Focuses more on video generation

Diffusion Models

Training

Learns p(x)

$\mathbf{x}$

$\mathbf{z}$

Model

$\hat{\mathbf{x}}$

Loss

Constructed by hiding information from x

Target

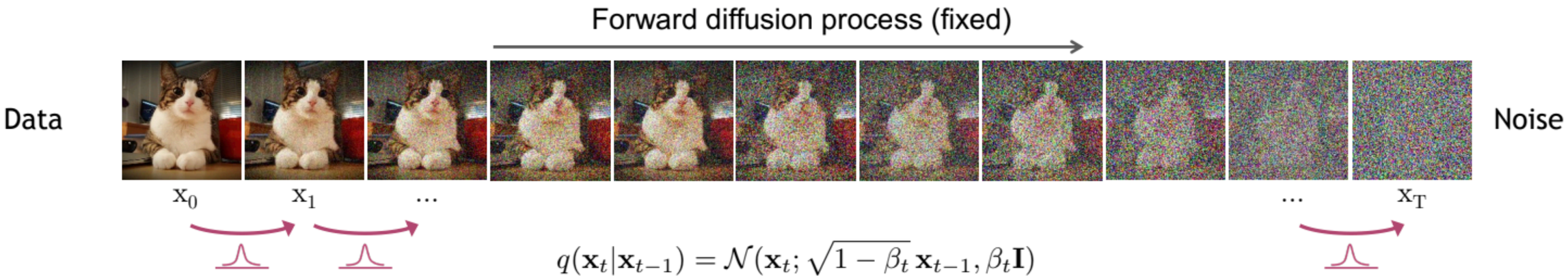
Information hiding is done by a Markovian noising process

Inference

Model

Sample from p(x)

Diffusion: Forward Process



Biggest assumption

$\mathbf{x}_T$ *Has NO information about the input*

Matters because “latent” is exactly of the same dimensionality as x

From https://cvpr2022-tutorial-diffusion-models.github.io/

Training

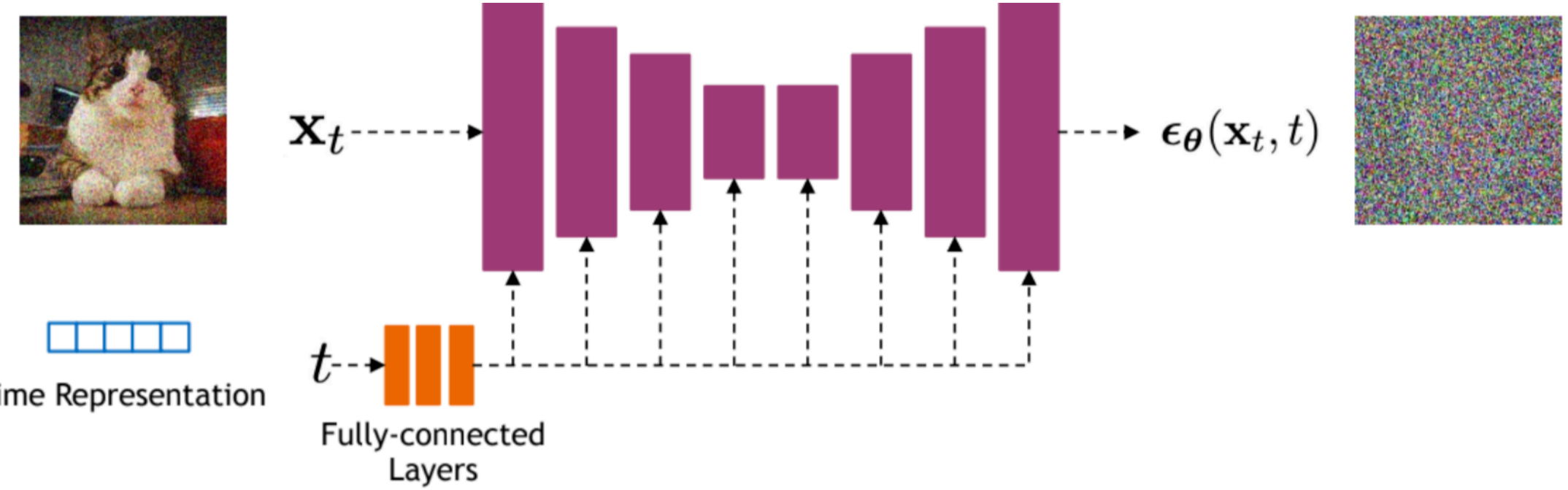
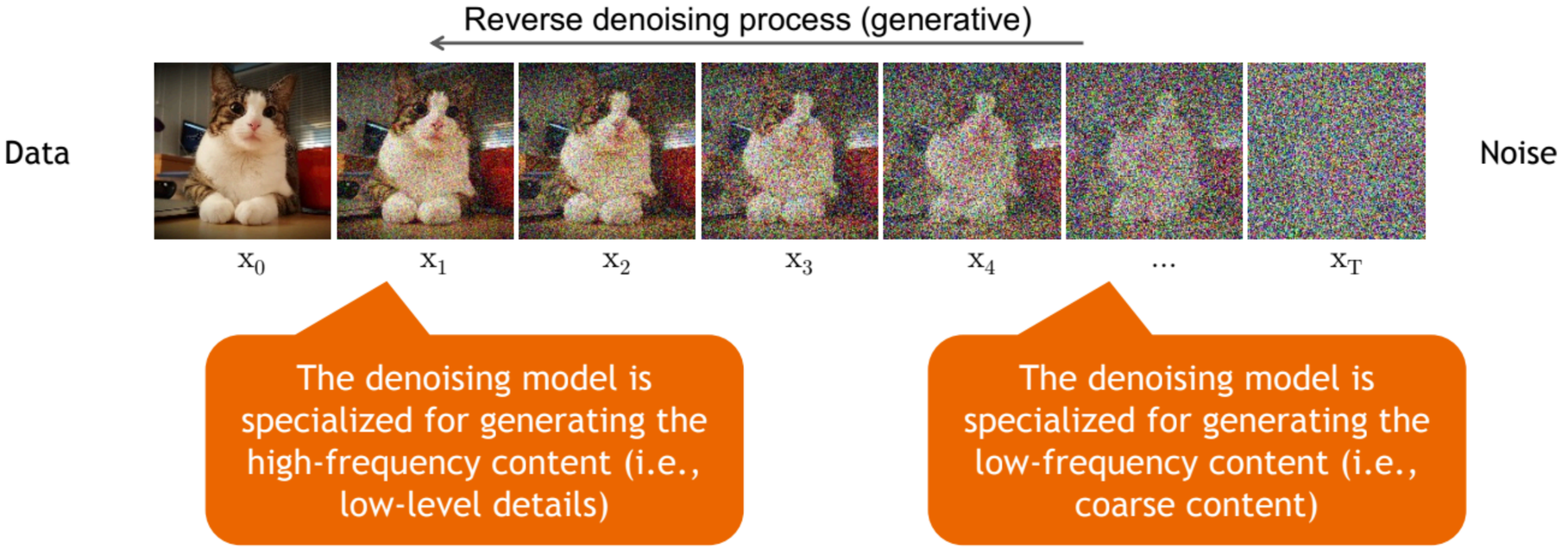


Image is noised by adding “epsilon” noise
Given the noisy image model is trained to predict “epsilon”

From https://cvpr2022-tutorial-diffusion-models.github.io/

Reverse denoising



From https://cvpr2022-tutorial-diffusion-models.github.io/

Reverse denoising

Algorithm 1 Training

1: **repeat**
2: $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
3: $t \sim \text{Uniform}(\{1, \dots, T\})$
4: $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
5: Take gradient descent step on
$$\nabla_\theta \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t) \right\|^2$$
6: **until** converged

Algorithm 2 Sampling

1: $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
2: **for** $t = T, \dots, 1$ **do**
3: $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
4: $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$
5: **end for**
6: **return** $\mathbf{x}_0$

From https://cvpr2022-tutorial-diffusion-models.github.io/

Text-to-Image Diffusion Models

$\mathbf{x}$

A tree on a beach

$\mathbf{c}$

Diffusion Model

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{c})$$

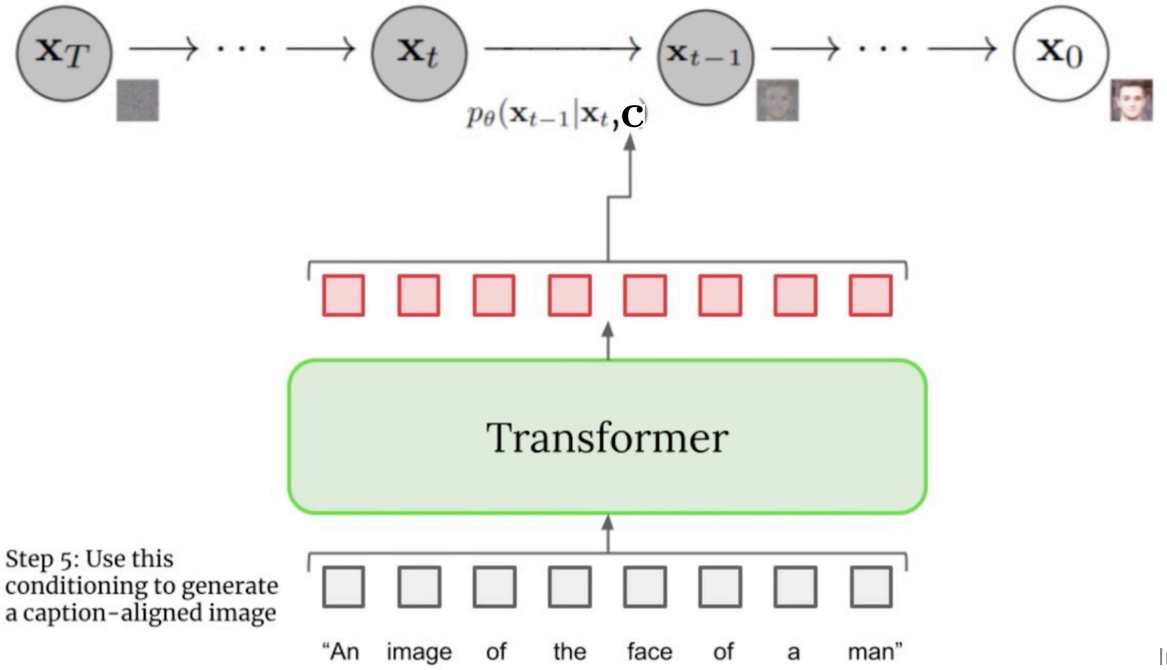


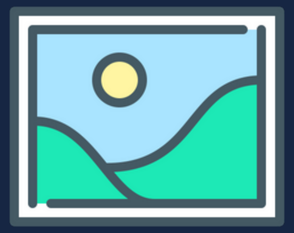
Image from https://eugeneyan.com/writing/text-to-image/

Design space for text-to-image models is HUGE

- Representing “x”
  - As RGB pixel values, e.g., DALLE-2, Imagen
  - Encode into a latent, e.g., StableDiffusion
- Encoder architecture
  - UNet with self-attention
  - Transformer
- Text-encoder
  - Encoder-LLM, e.g., T5
  - Joint-embedding models, e.g., CLIP
- Diffusion parametrization
  - Noise schedule
  - Exact way of predicting “epsilon” noise
  - Sampler

Overview

Evaluation



Controllable
Image Generation

Video Generation

Evaluation

Evaluations

- Measure two things
- **Image quality**
  - Distributional metrics
- **(Image, text) compatibility**
  - Similarity metrics

Image Quality

GLIDE - Dhariwal et al., 2021

Image Quality

- Cannot measure metrics such as SSIM or PSNR
  - Typically used for compression
  - These are defined wrt a “target” image

Compressed image

True image

- Rely on distributional metrics
  - Measure if the distribution of the generated and real images similar

Frechet Inception Distance

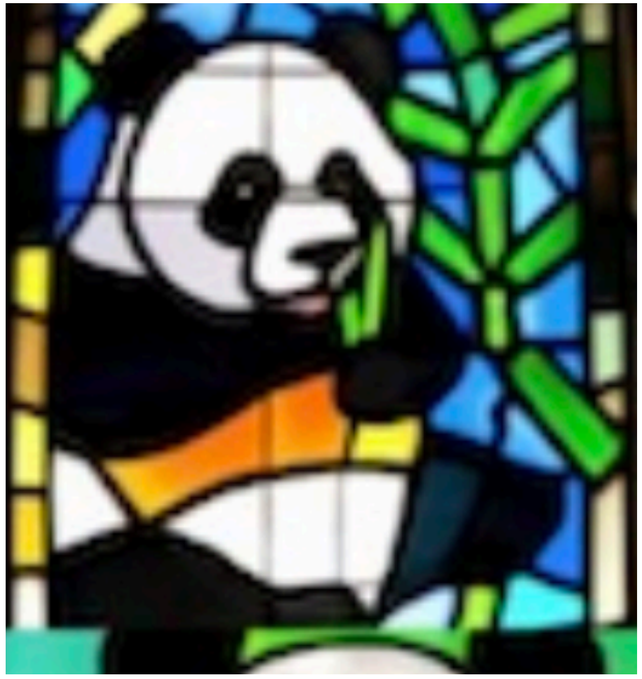
- Measure similarity of real/generated images using embeddings from an ImageNet pretrained Inception network

$$FID = ||\mu_X - \mu_Y||^2 - Tr(\sum_X + \sum_Y - 2\sqrt{\sum_X \sum_Y})$$

"GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium - Martin et al., 2017

(Image, text) compatibility

A panda painted on a stained window glass



(Image, text) compatibility

- Measured using similarity of image, text embeddings computed by CLIP

Open Questions in evaluation

- Are FID/FVD good metrics?
  - Models are trained on ImageNet or Kinetics
  - Old architectures
  - Use small datasets of “real” samples as a reference
- CLIP similarity
  - CLIP itself is a “bag of words” text encoder
  - Not good at many tasks (counting)

What is the “gold standard” evaluation?

- Human preferences!
  - Show images from “N” models, ask humans to pick the best one
  - Expensive but ultimately what matters for many applications

SelfEval: Leveraging the discriminative nature of generative models for evaluation

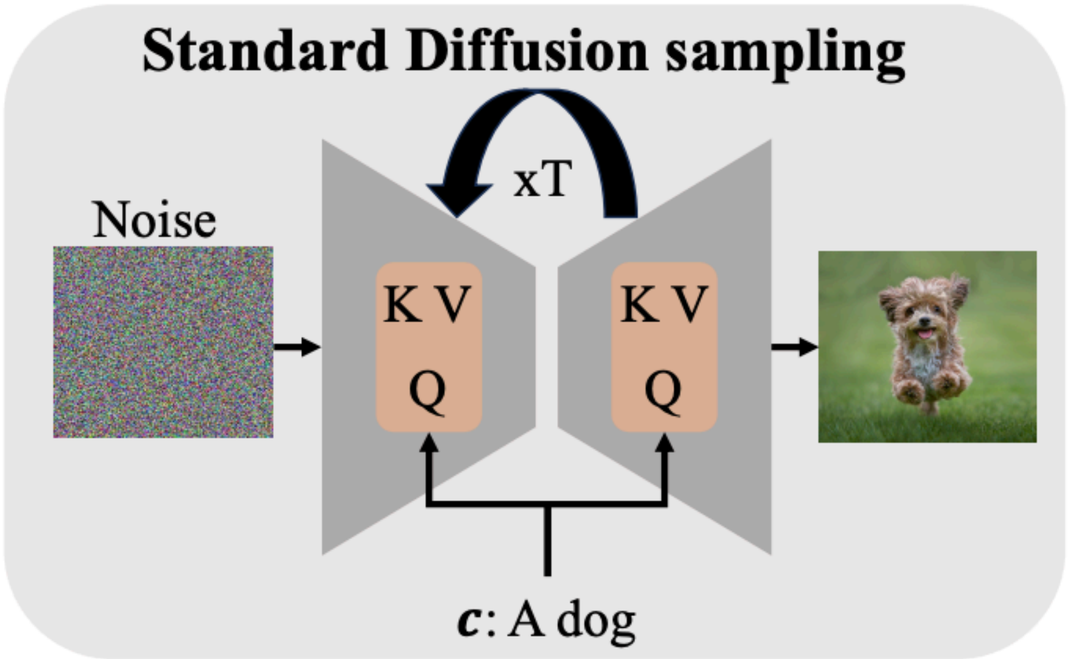
Sai Saketh Rambhatla, Ishan Misra

Background: Diffusion Models

Standard Diffusion Model (Reverse Process)

Estimate the likelihood of the sample “x”

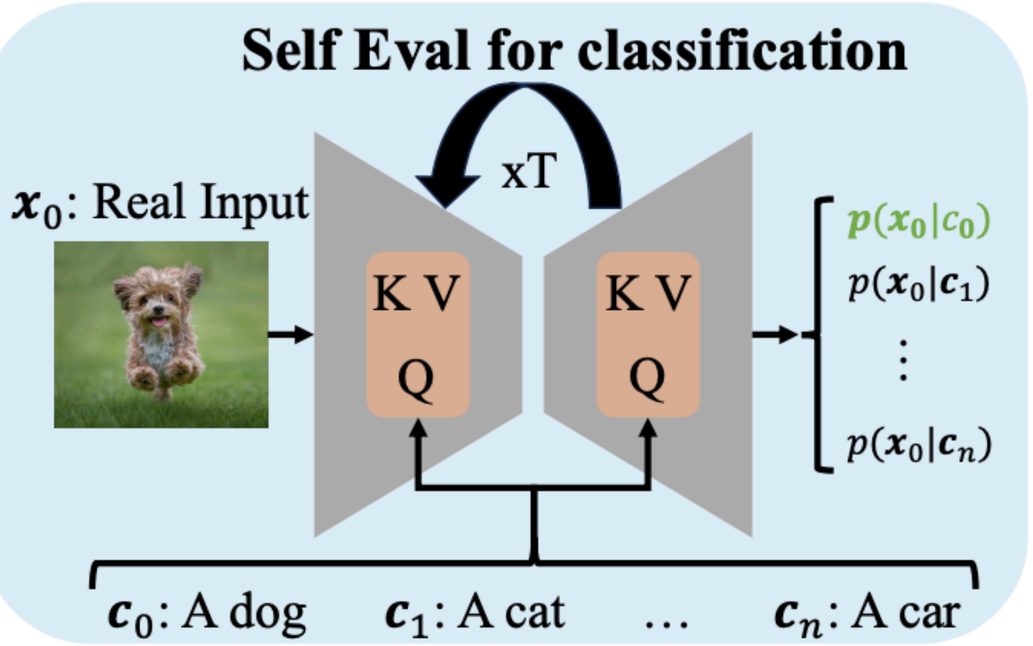
$$p_\theta(\mathbf{x}_0|\mathbf{c}) = \int p_\theta(\mathbf{x}_{0:T}|\mathbf{c}) d\mathbf{x}_{1:T}$$



SelfEval: Diffusion models as classifiers

Estimate the likelihood of the sample “x” for each of the “c” classes. Pick the one that maximizes the likelihood

$$p_\theta(\mathbf{x}_0|\mathbf{c}) = \int p_\theta(\mathbf{x}_{0:T}|\mathbf{c})d\mathbf{x}_{1:T}$$



We show numerically stable ways of estimating this likelihood

See also: Your diffusion model is secretly a classifier

So what can SelfEval do?

- Diffusion models for (image, text) understanding
- Use image understanding scores to benchmark diffusion models

Winoground Challenge

Diffusion models are comparable to SOTA discriminative models for this task

(a) some plants surrounding a lightbulb

(b) a lightbulb surrounding some plants

Table 1: Diffusion models evaluated on the Winoground dataset. We measure the image score (accuracy of picking the correct image given a text prompt) and text score (accuracy of picking the correct text given an image). Using SELFEVAL allows us to use diffusion models for both tasks unlike prior work Li et al. (2023a) which leads to zero image score.

Method	Model	Image Score	Text Score
CLIP (ViT-L/14)	–	8.00	30.25
OCLIP (ViT-H/14)	–	12.75	30.75
Li et al. (2023a)	LDM-CLIP	0	34.00
SELFEVAL	LDM-CLIP	7.25	22.75
SELFEVAL	PDM-CLIP	14.00	17.00
SELFEVAL	PDM-T5	12.00	28.25
SELFEVAL	LDM-T5	13.50	29.00

Winoground: Thrush et al., 2022

Evaluating diffusion models: problems with CLIP score

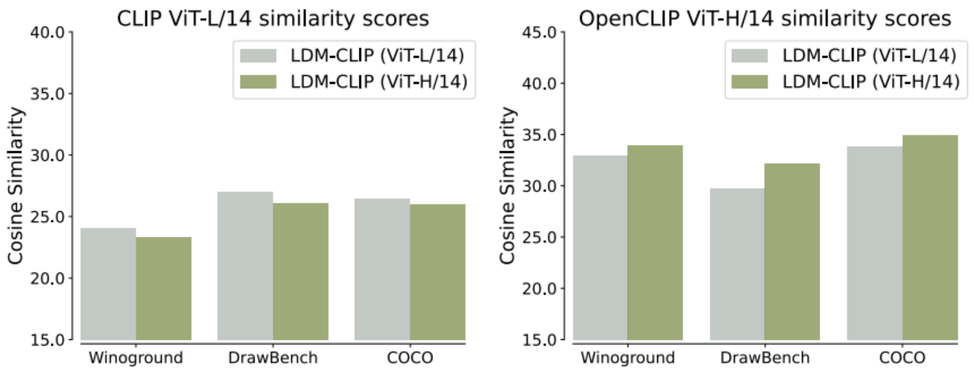


Table 2: Performance of CLIP on the benchmark. We evaluate the zero-shot performance of CLIP (ViT-L/14 visual backbone) on the six tasks. "Random" denotes the random chance accuracy. CLIP achieves impressive performance on the tasks of `Color` and `Shape`. We observe that the performance of CLIP is close to random on `Attribute binding`, `Count`, `Spatial` and `Text corruption`. This makes CLIP unsuitable for evaluating text faithfulness of generative models on prompts from these tasks.

Model	Attribute binding	Color	Count	Shape	Spatial	Text corruption
Random	50	25	25	33	25	20
CLIP	55.40	85.20	67.80	91.10	40.50	51.00

Which CLIP model to use?

Leads to different answers

CLIP model could be used in training the diffusion model

Limitations of CLIP leak into evaluations

CLIP is bad at many aspects such as counting, attributes, text corruption

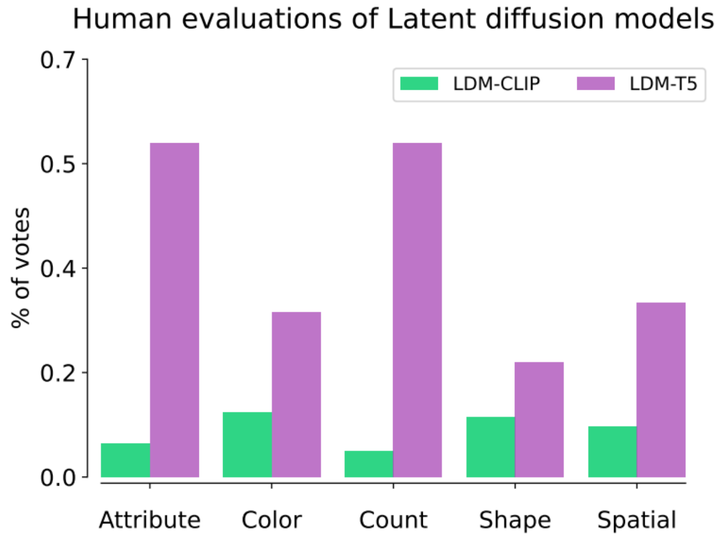
CLIP score is unreliable for these aspects

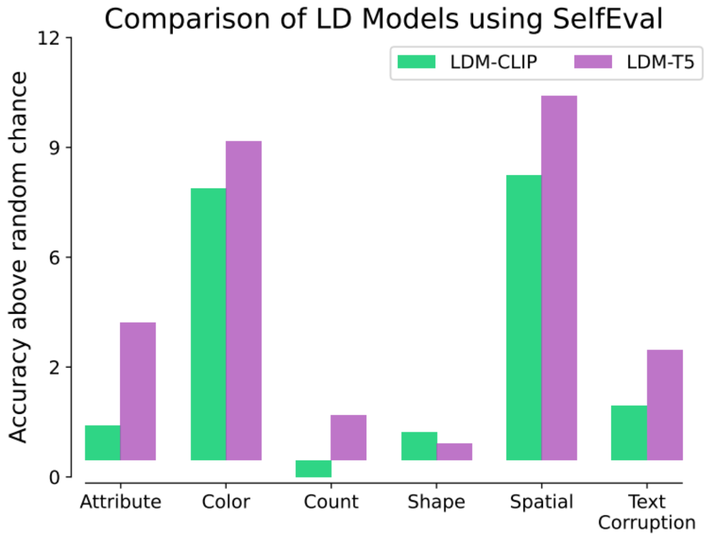
Evaluating diffusion models using SelfEval

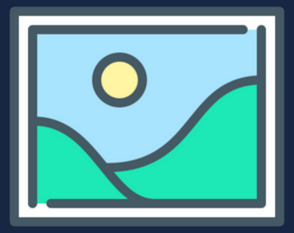
No need for external models like CLIP

Re-use image-understanding datasets

Agrees with human evaluations







Controllable
Image Generation

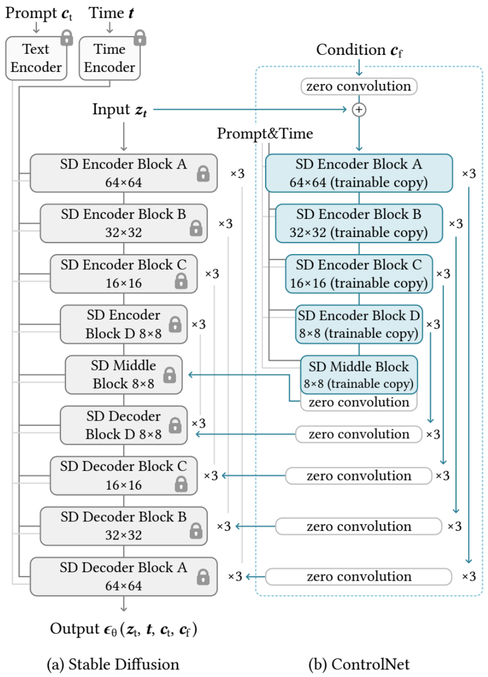
InstanceDiffusion: Instance-level control for image generation

XuDong Wang, Trevor Darrell, Sai Saketh Rambhatla, Rohit Girdhar, Ishan Misra

Available in Diffusers & ComfyUI!

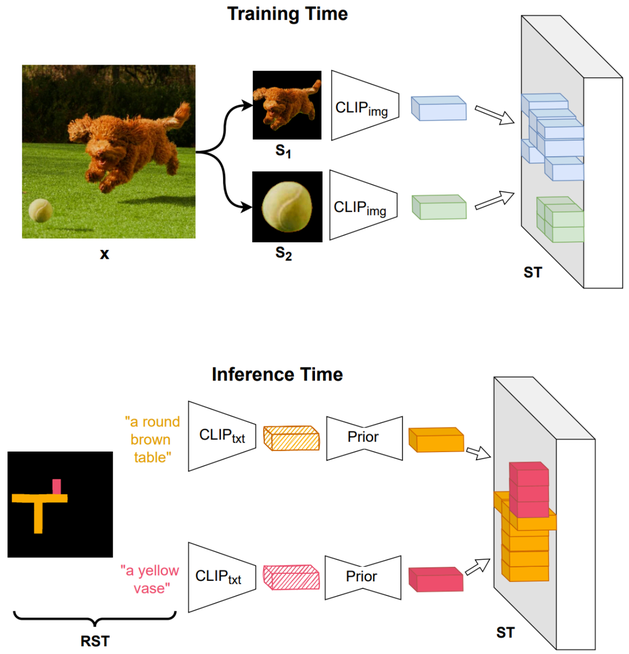
Adding more control to diffusion models

Only support semantic segmentation mask inputs and do not differentiate between instances of the same class



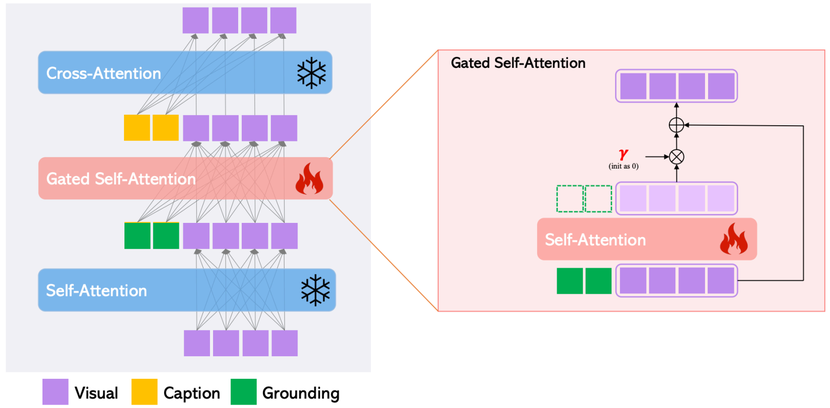
ControlNet

Zhang et. al. [ICCV 2023]



SpaText

Avrahami et. al. [CVPR 2023]



GLIGEN

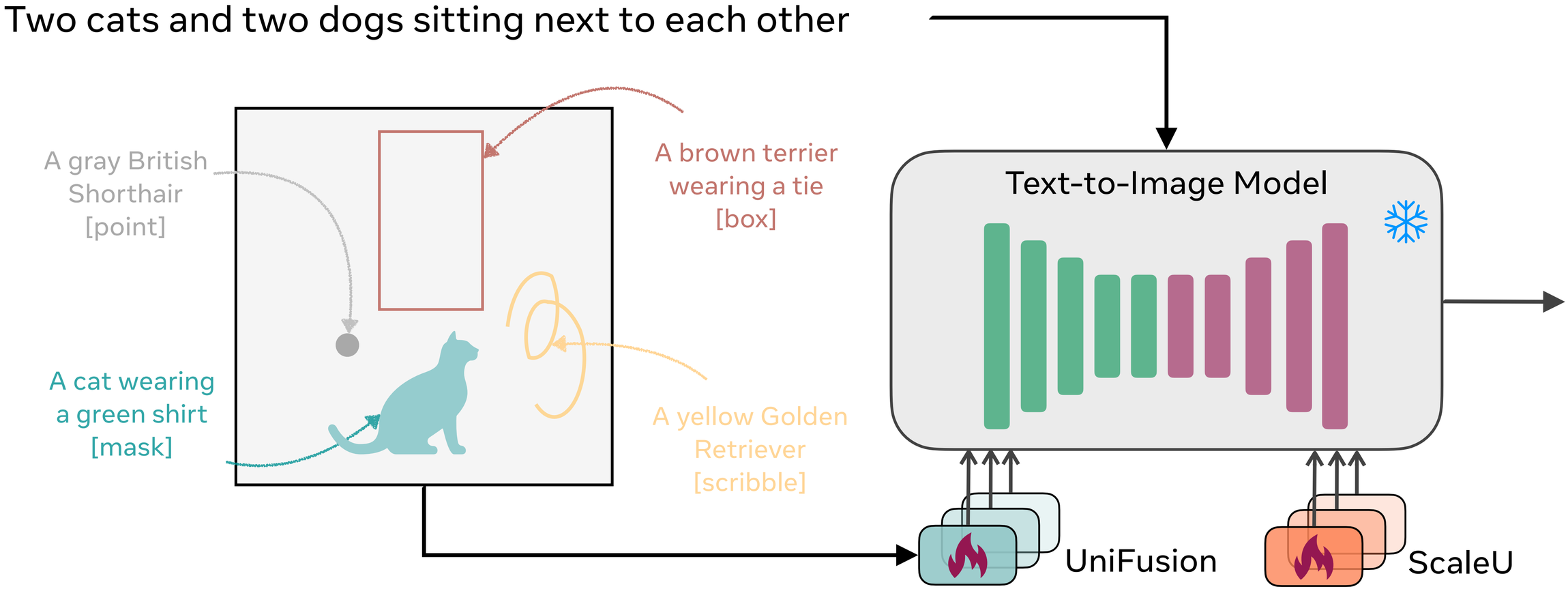
Li et. al. [CVPR 2023]

Key Idea

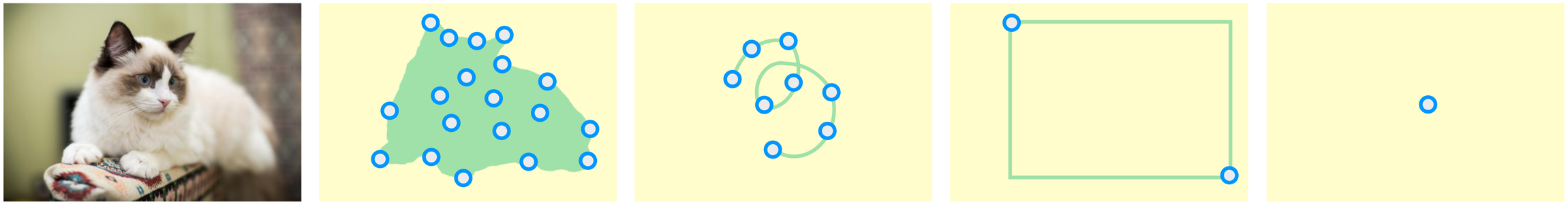
Add instance-level control to text-to-image model

Instance Diffusion

Allows for ***paired*** **instance-level prompts** and **their locations** (point, box, scribble or mask) to be specified

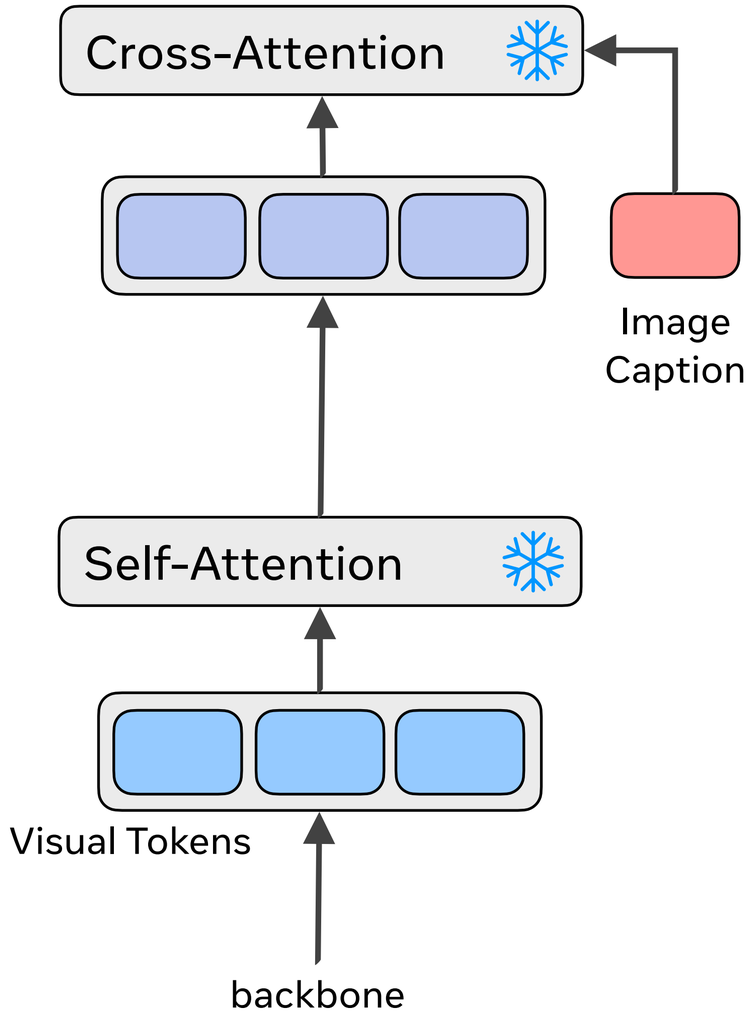


Different location formats as a set of points



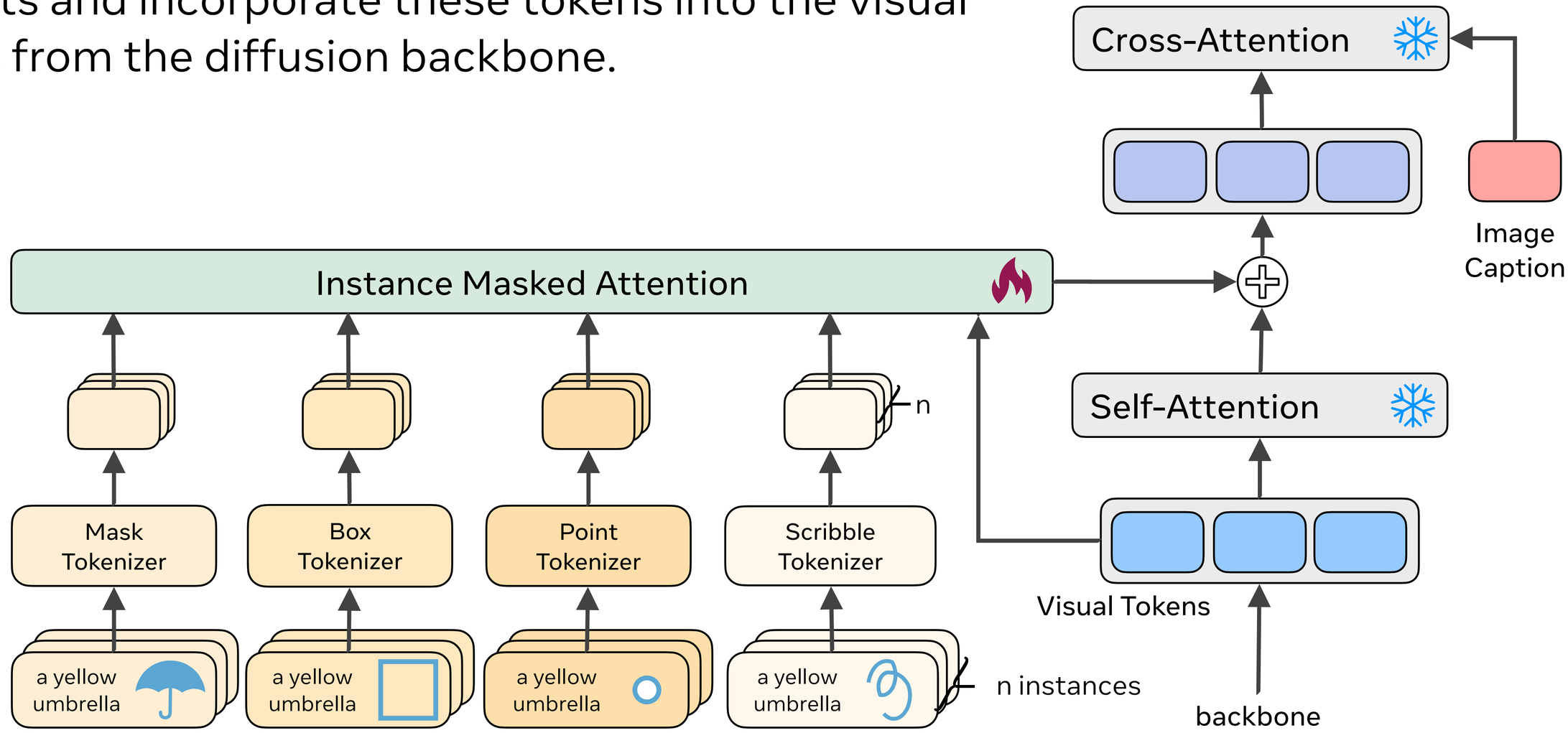
UniFusion: Fusing instance information with the T2I model

Transformer block of the Vanilla T2I Latent Diffusion Model (LDM)



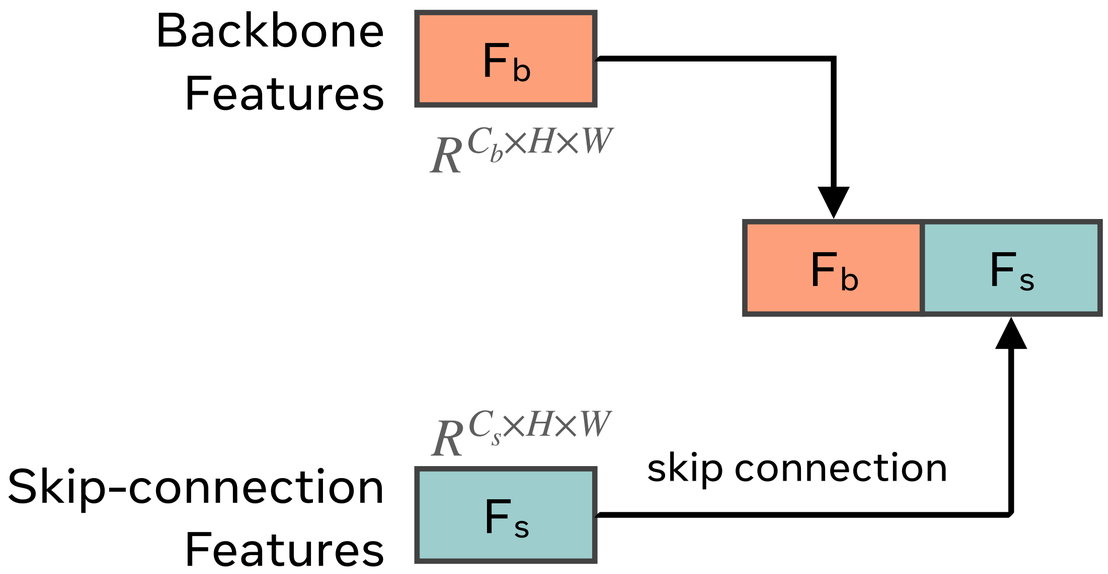
UniFusion: Fusing instance information with the T2I model

We tokenize various instance-level locations and text-prompts and incorporate these tokens into the visual tokens from the diffusion backbone.



ScaleU: Recalibrate the skip connection features in the UNet

UNet's Vanilla Skip Connection: Concatenating these two features directly leads to the network overlooking the semantic content of the main features [1]

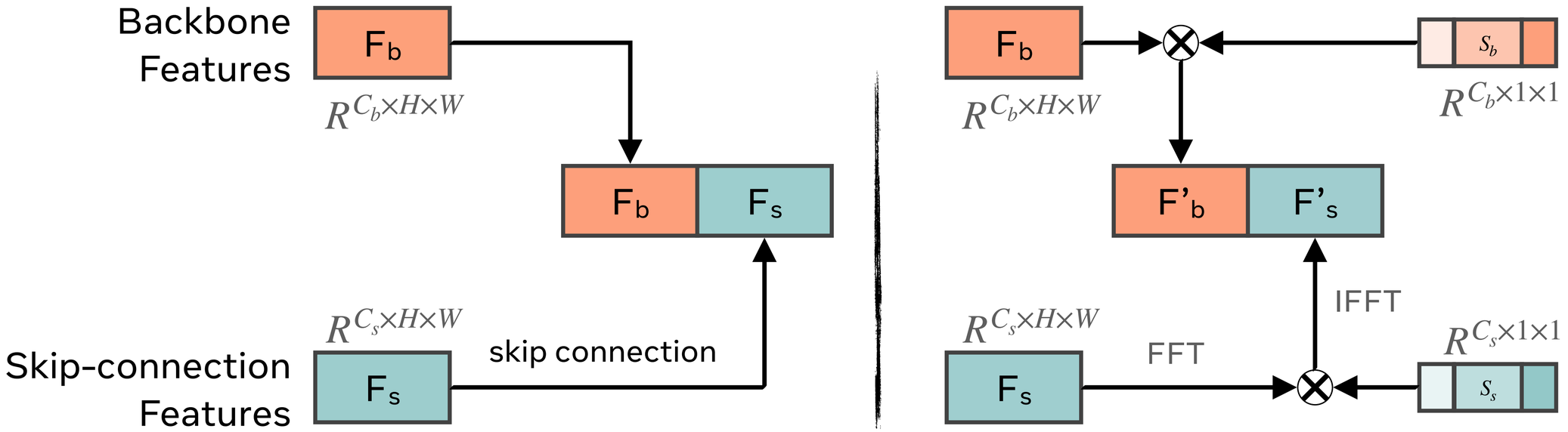


a) Skip Connection in UNet

[1] Chenyang Si et. al. Free lunch in diffusion u-net. arXiv preprint arXiv:2309.11497

ScaleU: Recalibrate the skip connection features in the UNet

ScaleU uses channel-wise and learnable vectors to dynamically recalibrate main feature and the low-frequency component of the skip connection feature.

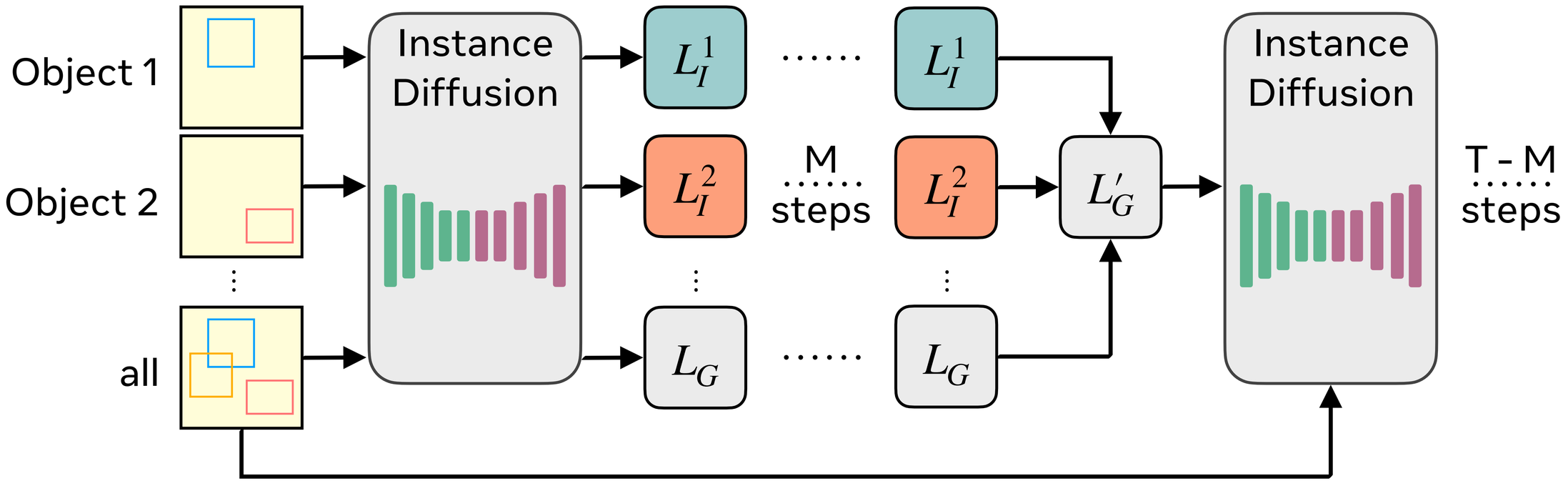


a) Skip Connection in UNet

b) Skip Connection w/ ScaleU

Multi Instance Sampler

Reduce information leakage across multiple instance conditionings



Example: Diverse locations and attributes

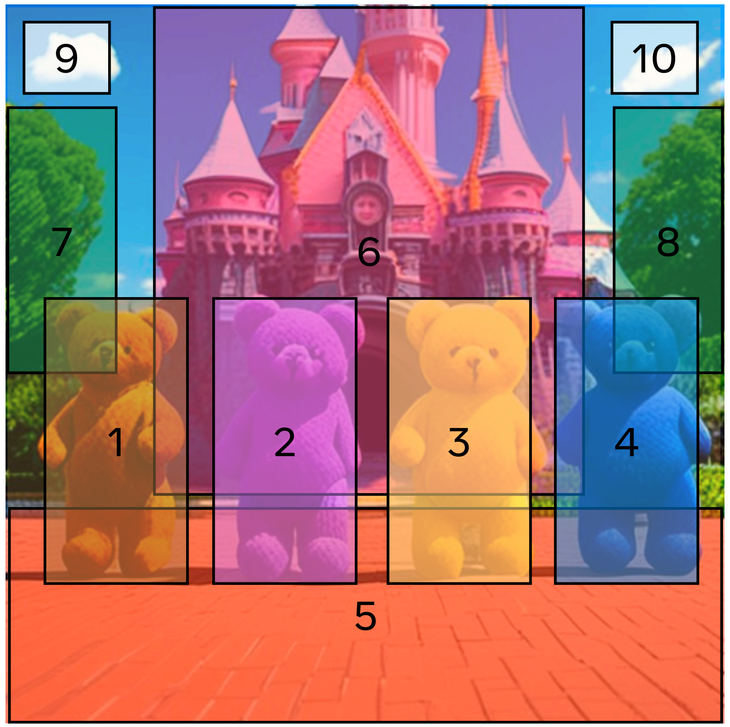


Image Caption: An image depicting in the morning. A ***brown*** cute teddy bear, a ***purple*** cute teddy bear, a ***yellow*** cute teddy bear, a ***blue*** cute teddy bear all standing side by side on a ***red*** brick road. The scene should be set in front of ***Pink*** Castle with clear ***blue*** sky overhead, punctuated by fluffy ***white*** clouds, and trees with ***green*** leaves. The ***pink*** castle should loom majestically in the background.

Instance Captions: 1-4) A ***brown/purple/yellow/blue*** teddy bear; 5) a ***red*** brick road; 6) ***Pink*** Castle; 7-8) ***green*** leaves; 9-10) fluffy ***white*** clouds

Example: Dense small objects

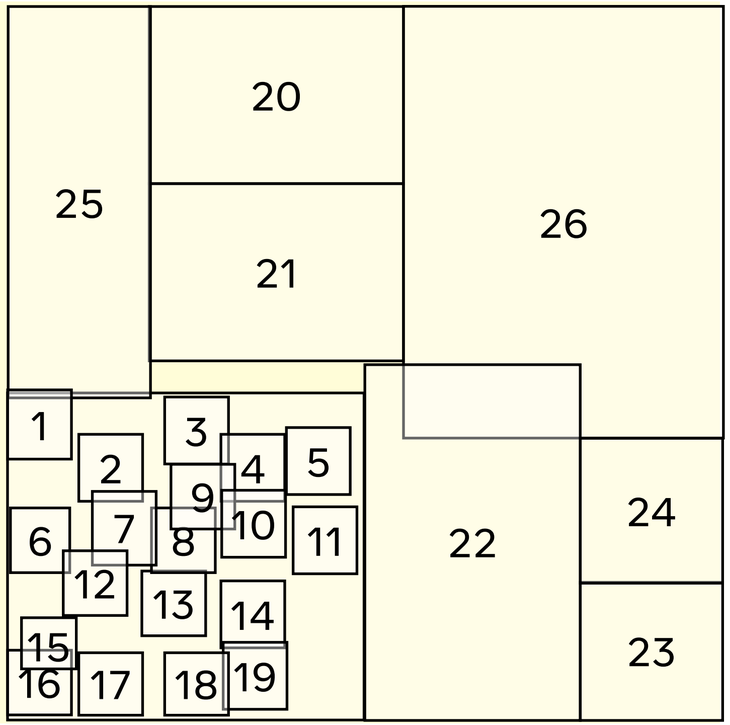
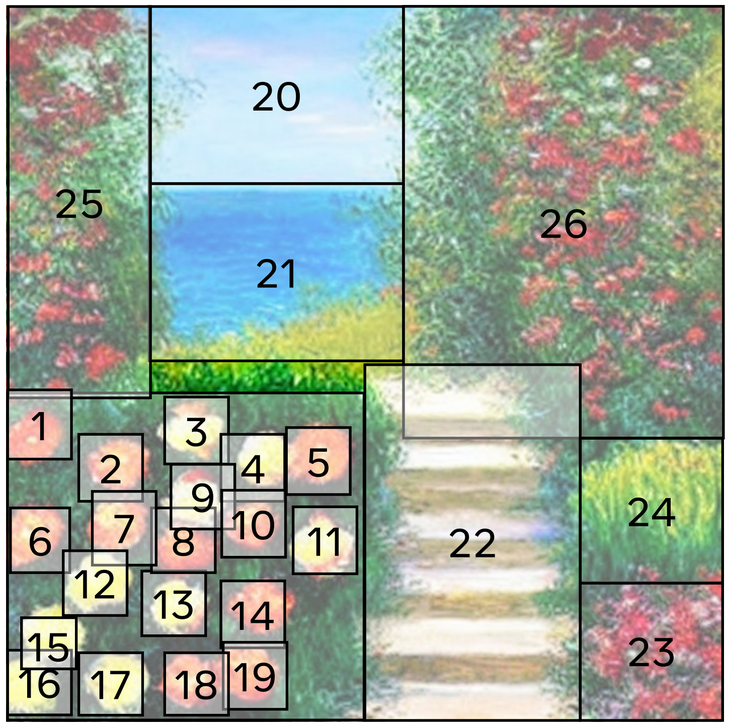


Image Caption: Craft an oil painting: Picture a seaside garden drenched in radiant hues of roses, lilies, and lavender, transitioning gracefully into the expansive azure ocean and blue sky. Integrate a weathered, rustic pathway with steps that invite viewers towards the water's edge, complemented by a prominent bouquet of flowers and plants.
Instance Captions: 1-19) roses; 20) sky; 21) ocean; 22) pathway with steps; 23) bouquet of flowers; 24) plant; 25-26) plants

Example: Dense small objects



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Instance Captions: 1-19) roses; 20) sky; 21) ocean; 22) pathway with steps; 23) bouquet of flowers; 24) plant; 25-26) plants

Example: Using a combination of instance-level inputs

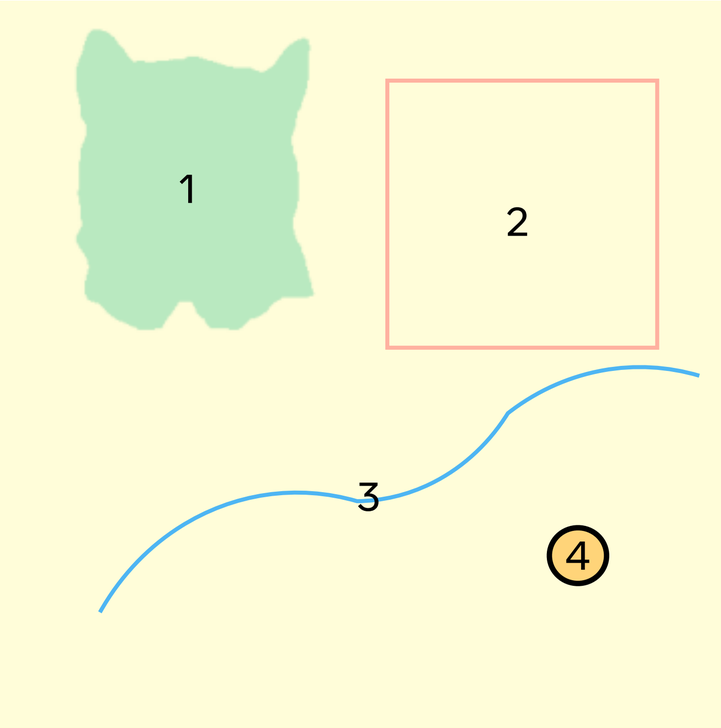
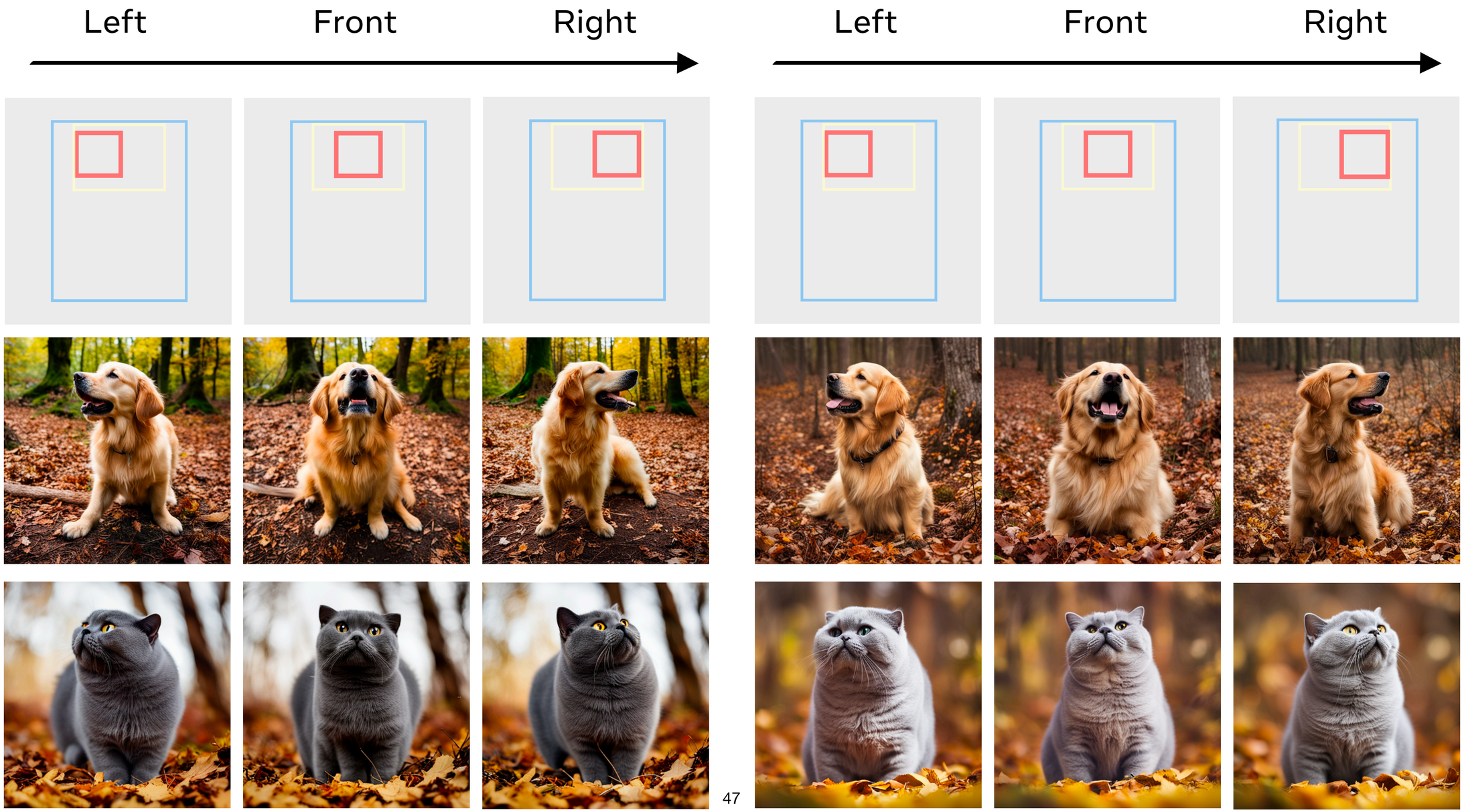


Image Caption: An image of two little husky puppy in a wicker basket.
Instance Captions: 1) a husky puppy sitting in a wicker basket + ***Mask.*** 2) a black and white husky puppy in a blue towel + ***Box***. 3) two husky puppies sitting in a wicker basket + ***Scribble***. 4) a blue towel + ***Point***

Example: Turning everyone's heads



Multi-round image generation

Image Caption: A cup of tea with tangerines, bananas, and cookies on the table. high quality. professional photo.
Instance Captions: 1) a cup of tea on a lace doily 2) a close up of three oranges on a black background 3) oranges in a glass bowl on a table 4) a tray of pastries on a table with oranges 5) a close up of some cookies on a table 6) oranges in a glass bowl 7) oranges in a glass bowl 8) an orange that has been cut in half on a table 9) an orange is cut in half 10) bananas 11) a bouquet of flowers on a table 12) a donut / lighted candle

Multi-round image generation

Add *“a bouquet of flowers”* and *“A donut”*

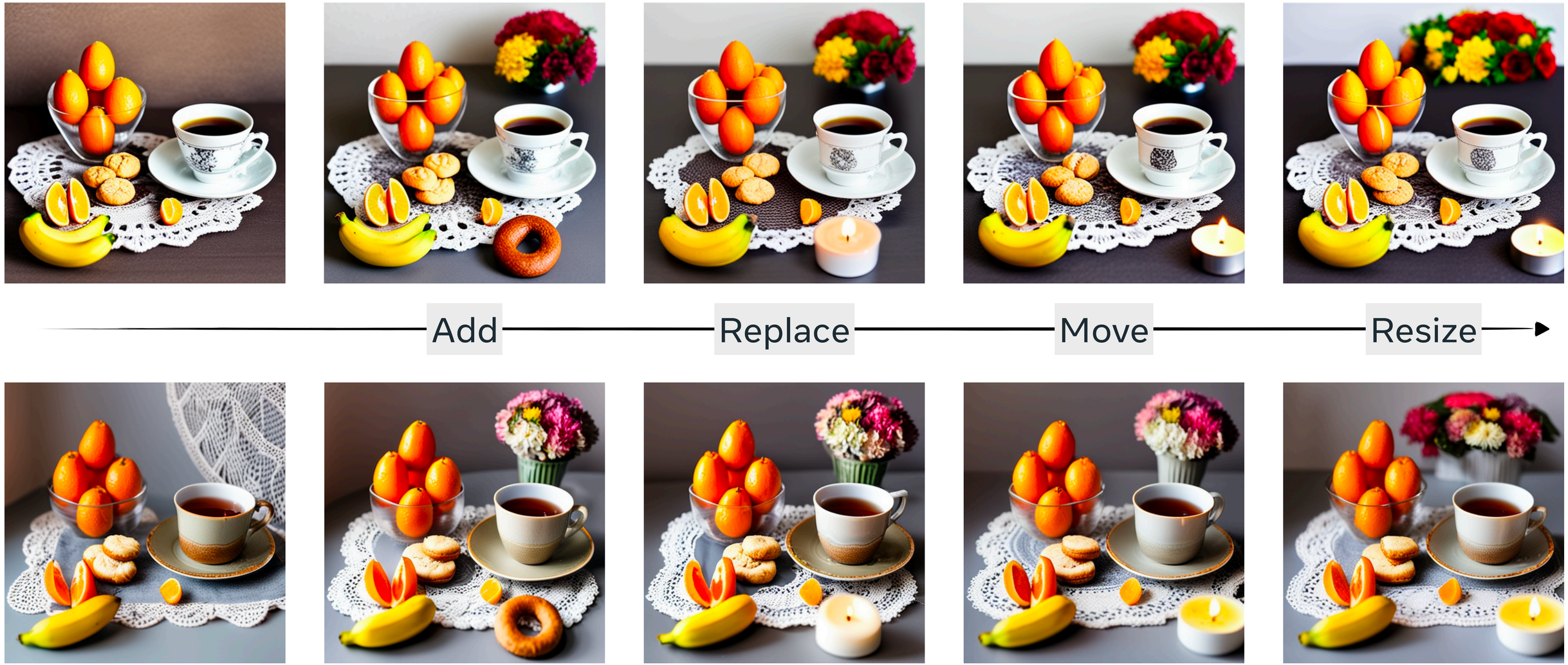
Image Caption: A cup of tea with tangerines, bananas, and cookies on the table. high quality. professional photo.
Instance Captions: 1) a cup of tea on a lace doily 2) a close up of three oranges on a black background 3) oranges in a glass bowl on a table 4) a tray of pastries on a table with oranges 5) a close up of some cookies on a table 6) oranges in a glass bowl 7) oranges in a glass bowl 8) an orange that has been cut in half on a table 9) an orange is cut in half 10) bananas 11) a bouquet of flowers on a table 12) a donut / lighted candle

Multi-round image generation

Replace *"A donut"* to *"A lighted candle"*

Image Caption: A cup of tea with tangerines, bananas, and cookies on the table. high quality. professional photo.
Instance Captions: 1) a cup of tea on a lace doily 2) a close up of three oranges on a black background 3) oranges in a glass bowl on a table 4) a tray of pastries on a table with oranges 5) a close up of some cookies on a table 6) oranges in a glass bowl 7) oranges in a glass bowl 8) an orange that has been cut in half on a table 9) an orange is cut in half 10) bananas 11) a bouquet of flowers on a table 12) a donut / lighted candle

Multi-round image generation



vs. ImageEdit: We don’t need to be conditioned on previously generated images.

Quantitative Eval: Location/Shape Conditions

InstanceDiffusion substantially outperforms prior work on ALL metrics!

Location format input →	Boxes				Instance Masks					Points		Scribble	
Method	AP^{box}	AP^{box}_{50}	AR^{box}	FID (↓)	IoU	AP^{mask}	AP^{mask}_{50}	AR^{mask}	FID (↓)	PiM	FID (↓)	PiM	FID (↓)
Upper bound (real images)	50.2	66.7	61.0	-	-	40.8	63.5	58.0	-	-	-	-	-
GLIGEN [33]	19.6	35.0	30.7	27.0	-	-	-	-	-	-	-	-	-
GLIGEN [33]*	19.3	34.6	31.1	-	-	-	-	-	-	-	-	30.2†	32.4†
ControlNet [63]‡	-	-	-	-	-	6.5	13.8	12.9	-	-	-	-	-
DenseDiffusion [27]	-	-	-	-	35.0 / 48.6	-	-	-	-	-	-	-	-
SpaText [4]‡	-	-	-	-	-	5.3	12.1	10.7	-	-	-	-	-
InstanceDiffusion	**39.0**	**55.4**	**52.4**	**23.1**	**61.6 / 71.4**	**23.5**	**46.1**	**34.9**	**25.2**	**79.7**	**27.1**	**67.3**	**26.4**
vs. prev. SoTA	**+19.4**	**+20.4**	**+21.3**	**-3.9**	**+25.4 / +22.8**	**+17.0**	**+32.3**	**+22.0**	-	-	-	**+37.1**	**-6.0**

Video Generation

Text-to-video generation

Input: Text prompt describing your favorite things

Output: Video (related to the text prompt)

A brief timeline of video generation models

SyncDraw
Aug 2017

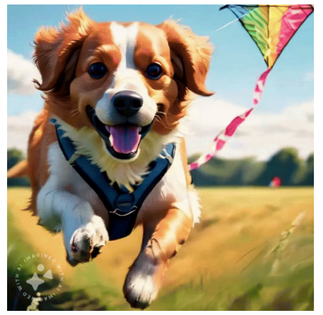
VideoGPT
Apr 2021

VDM
Apr 2022

Meta Make-A-Video
Sep 2022

Google Imagen Video
Google Phenaki
Pika Labs
RunwayML Gen2

Meta Emu Video
Nov 2023

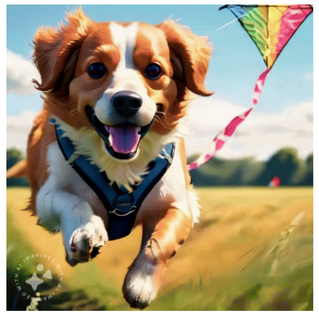


OpenAI SORA
Feb 2024

Image/Video sources from respective papers

Google Imagen Video
Google Phenaki
Pika Labs
RunwayML Gen2

Meta Emu Video
Nov 2023



OpenAI SORA
Feb 2024

Gen3 Alpha
LumaLabs
Kling

Meta Movie Gen
Oct 2024

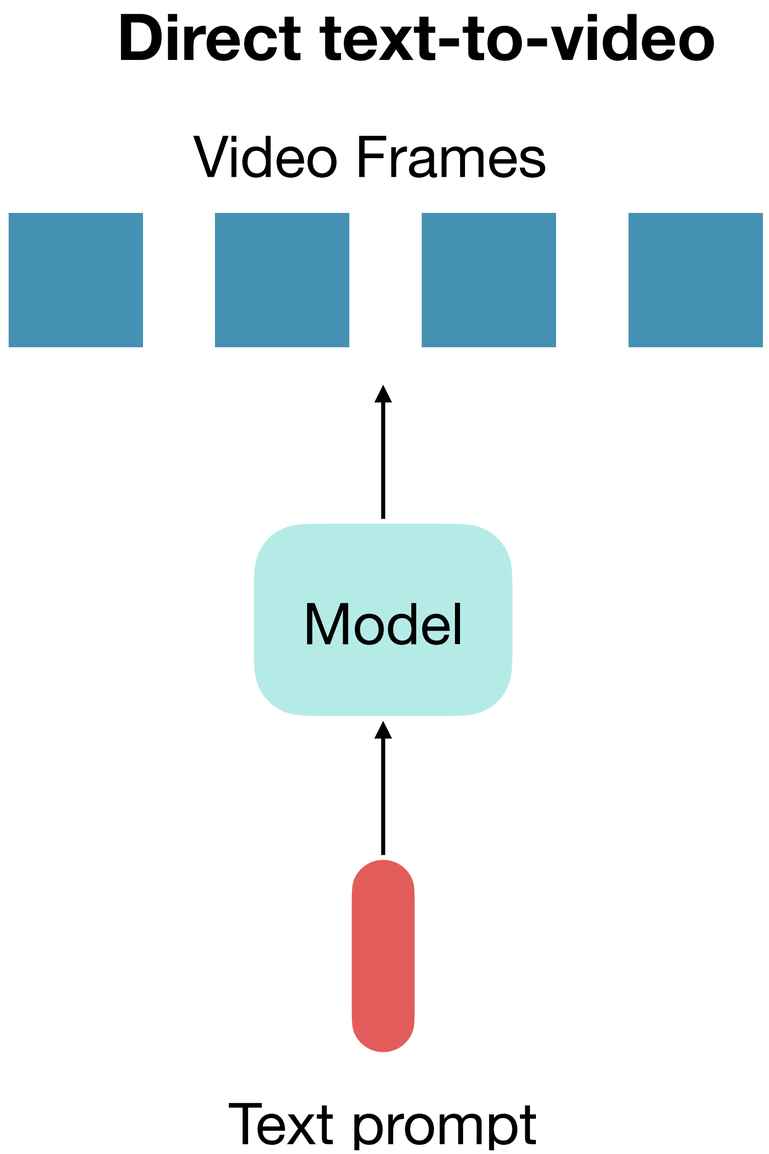
Movie Gen
Video Personalize

Movie Gen Video

Movie Gen
Video Edit

Image/Video sources from respective papers

Most text-to-video approaches



- Directly generate “N” frames of a video conditioned on the text prompt
- Deep cascades of models of produce high frame-rate high resolution videos

Imagen & Imagen video

https://imagen.research.google/video/

Imagen

- Text-to-image model
- Foundation for Imagen-video
- Operates in RGB pixel space

Data

- Image-text pairs
- LAION-400M
- Internal (~500M images)

Model

- Diffusion models
- Cascading super-res
- Frozen text encoders

Sampler

- Classifier-free guidance
- Maximizing text-alignment

Scaling up

Image from:http://cs231n.stanford.edu/slides/2023/lecture_15.pdf
Imagen - Saharia et al., 2022

Imagen

A photo of a group of teddy bears eating pizza in Times Square.

Frozen text encoder

Embeddings

Embeddings +

64x64

Embeddings +

256x256

Embeddings +

1024x1024

Image from:http://cs231n.stanford.edu/slides/2023/lecture_15.pdf

Classifier Guidance

- Assume pairs of data (x, c). A classifier guidance diffusion model consists of
  - A trained conditional diffusion model
  - A trained classifier model on noisy data $\mathbf{x}_t$
- During sampling, at each denoising step, modify the score function to

$$\nabla_{\mathbf{x}_t} \log \tilde{p}_{\theta,\phi}(\mathbf{x}_t|\mathbf{c}) = \nabla_{\mathbf{x}_t} \log p_\theta(\mathbf{x}_t|\mathbf{c}) + \omega \nabla_{\mathbf{x}_t} \log p_\phi(\mathbf{c}|\mathbf{x}_t).$$

From the conditional diffusion model

From the classifier model

- Upweight samples that the classifier assigns high probability with, better alignment with c.
- Cons: need to train an additional classifier. Increase model complexity.

Image from:http://cs231n.stanford.edu/slides/2023/lecture_15.pdf

Classifier guidance: Dhariwal & Nichols, 2021

Classifier Guidance

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Image from:http://cs231n.stanford.edu/slides/2023/lecture_15.pdf

Classifier guidance: Dhariwal & Nichols, 2021

Classifier Guidance

Figure 3: Samples from an unconditional diffusion model with classifier guidance to condition on the class "Pembroke Welsh corgi". Using classifier scale 1.0 (left; FID: 33.0) does not produce convincing samples in this class, whereas classifier scale 10.0 (right; FID: 12.0) produces much more class-consistent images.

Image from: Classifier guidance: Dhariwal & Nichols, 2021

Classifier-free Guidance (CFG)

- Assume there're two diffusion models, one conditional model and one unconditional model.
- By Bayes' rule we can define an implicit classifier

$$p_\theta(\mathbf{c}|\mathbf{x}_t) \propto p_\theta(\mathbf{x}_t|\mathbf{c})/p_\theta(\mathbf{x}_t).$$

- The modified score function during sampling then becomes

$$\nabla_{\mathbf{x}_t} \log \tilde{p}_\theta(\mathbf{x}_t|\mathbf{c}) = (1+\omega)\nabla_{\mathbf{x}_t} \log p_\theta(\mathbf{x}_t|\mathbf{c}) - \omega\nabla_{\mathbf{x}_t} \log p_\theta(\mathbf{x}_t).$$

From the conditional diffusion model

From the unconditional diffusion model

- The two models can share weights, with the unconditional model taking a null class label c.

Image from:http://cs231n.stanford.edu/slides/2023/lecture_15.pdf

Classifier-free guidance: Ho & Salimans, 2021

Classifier-free Guidance (CFG)

- At training time
  - Dropout the conditioning
  - Helps the model behave like an unconditional model

$$\nabla_{\mathbf{x}_t} \log \tilde{p}_\theta(\mathbf{x}_t|\mathbf{c}) = (1+\omega)\nabla_{\mathbf{x}_t} \log p_\theta(\mathbf{x}_t|\mathbf{c}) - \omega\nabla_{\mathbf{x}_t} \log p_\theta(\mathbf{x}_t).$$

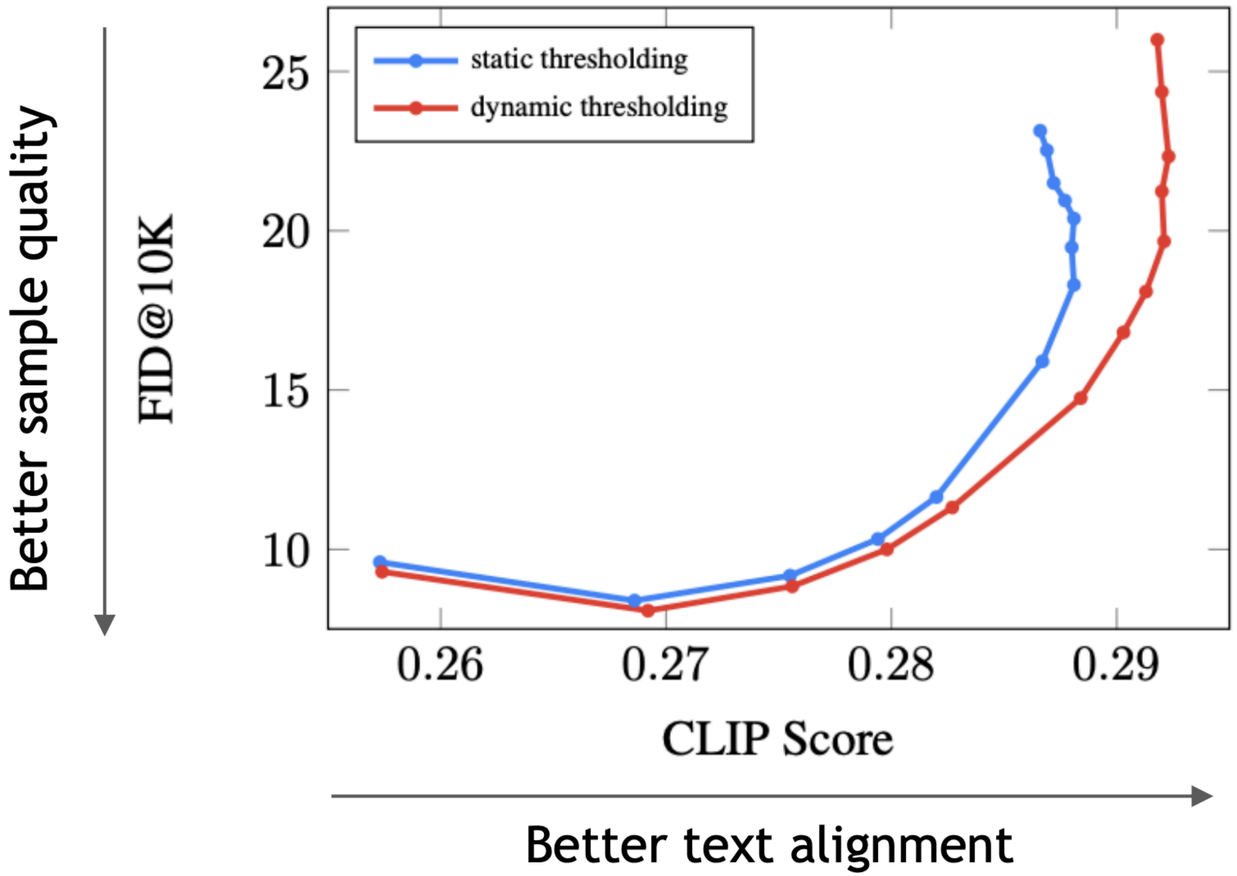
Classifier free guidance scale

Image from:http://cs231n.stanford.edu/slides/2023/lecture_15.pdf

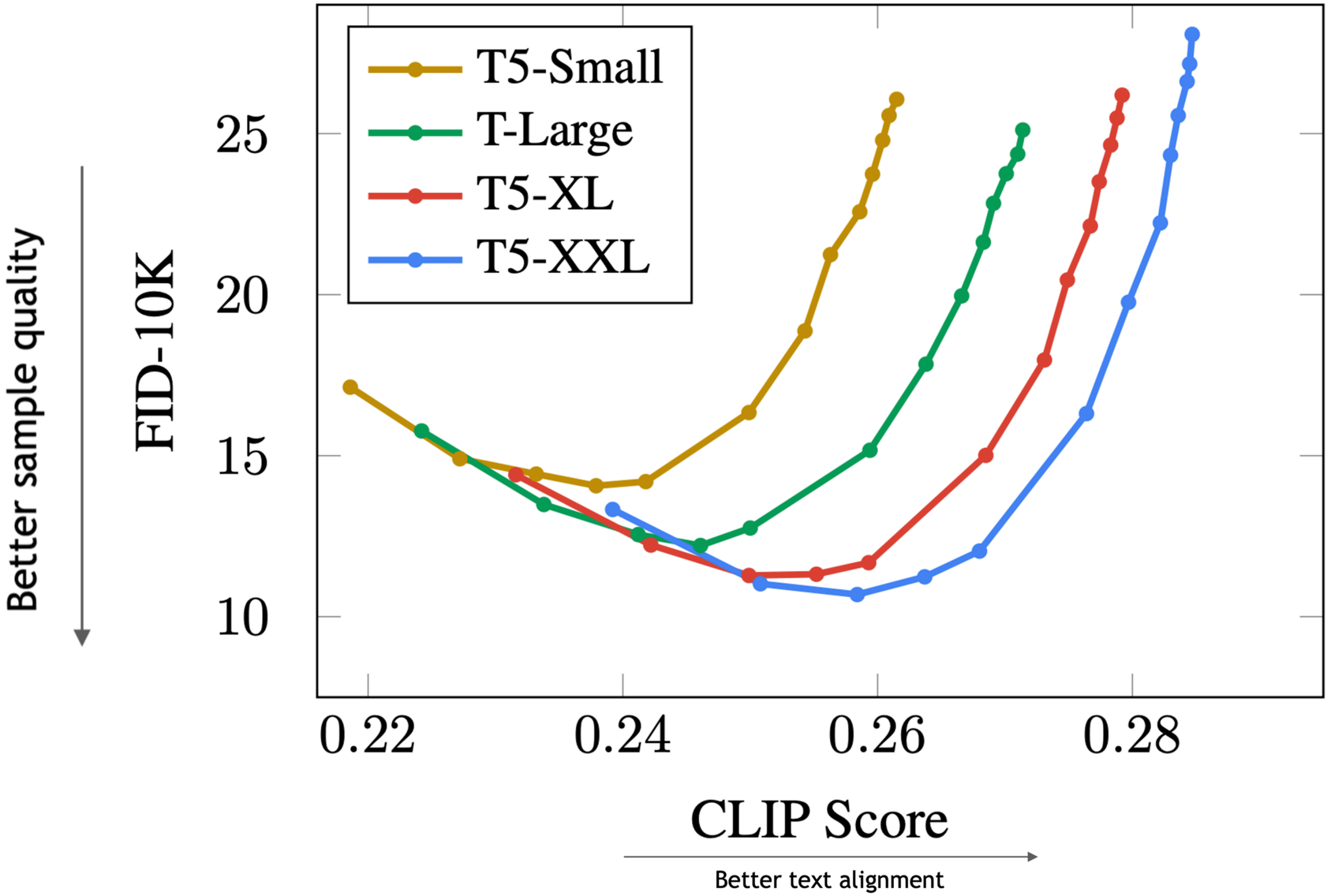
Classifier-free guidance: Ho & Salimans, 2021

Classifier-free Guidance (CFG)

- Trade-off between respecting the input text prompt & image quality

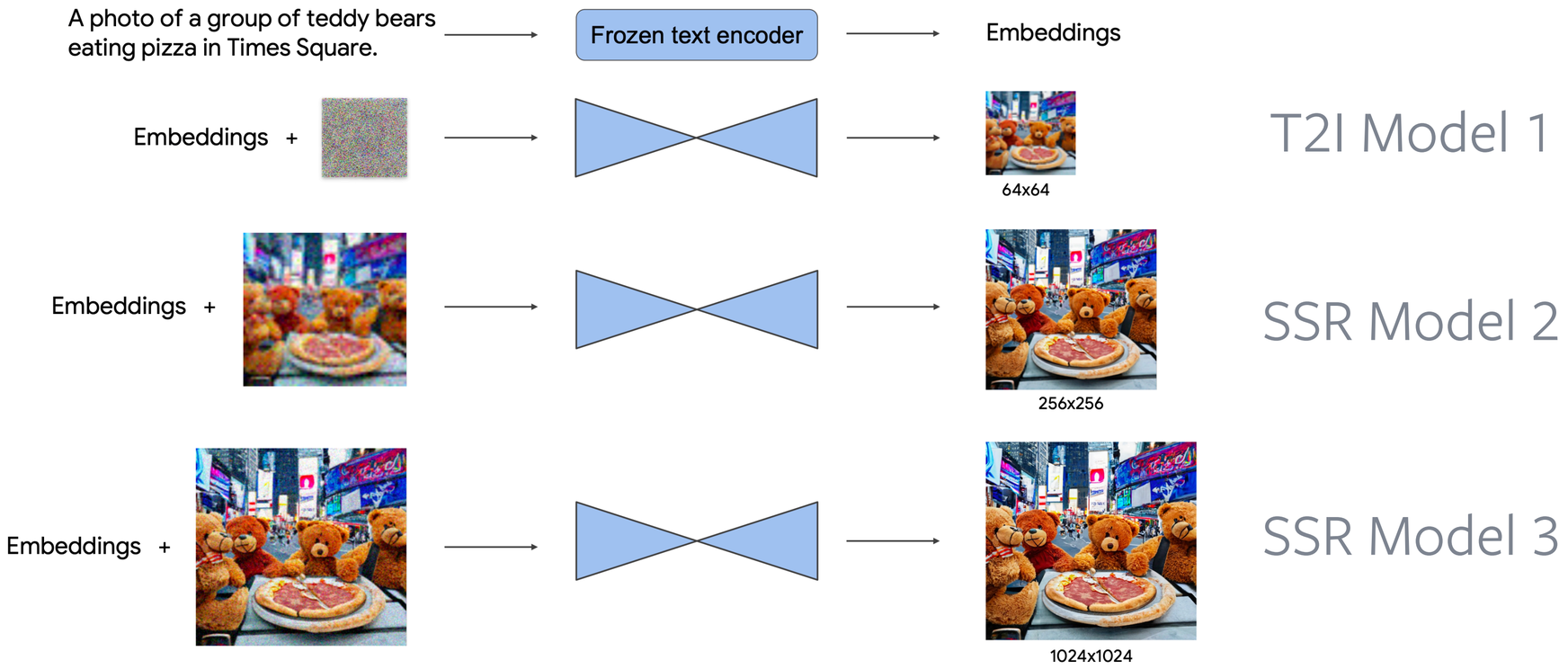


Text-encoder matters



T5-XXL has 4.6B parameters

An issue with cascades



In theory, cascaded models should correct errors produced in the previous stage

Reality - SSR models are trained with real images
At inference, they are fed generated images

Training super-res models

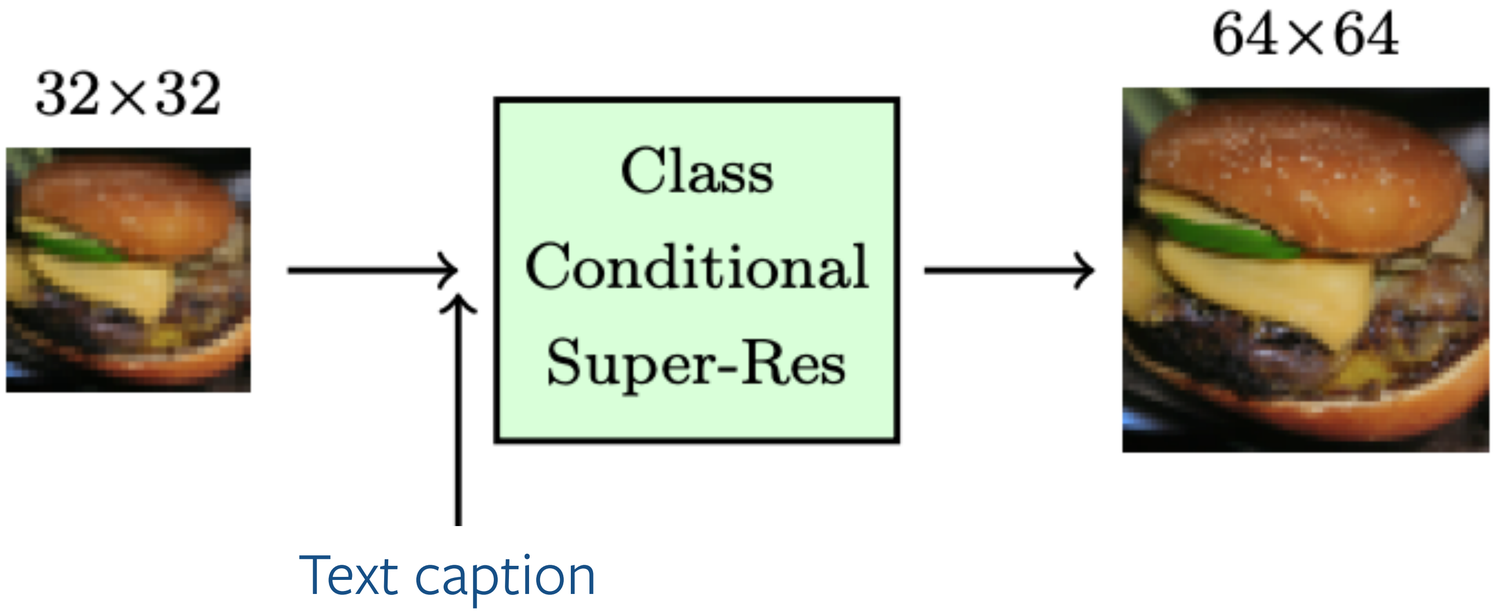
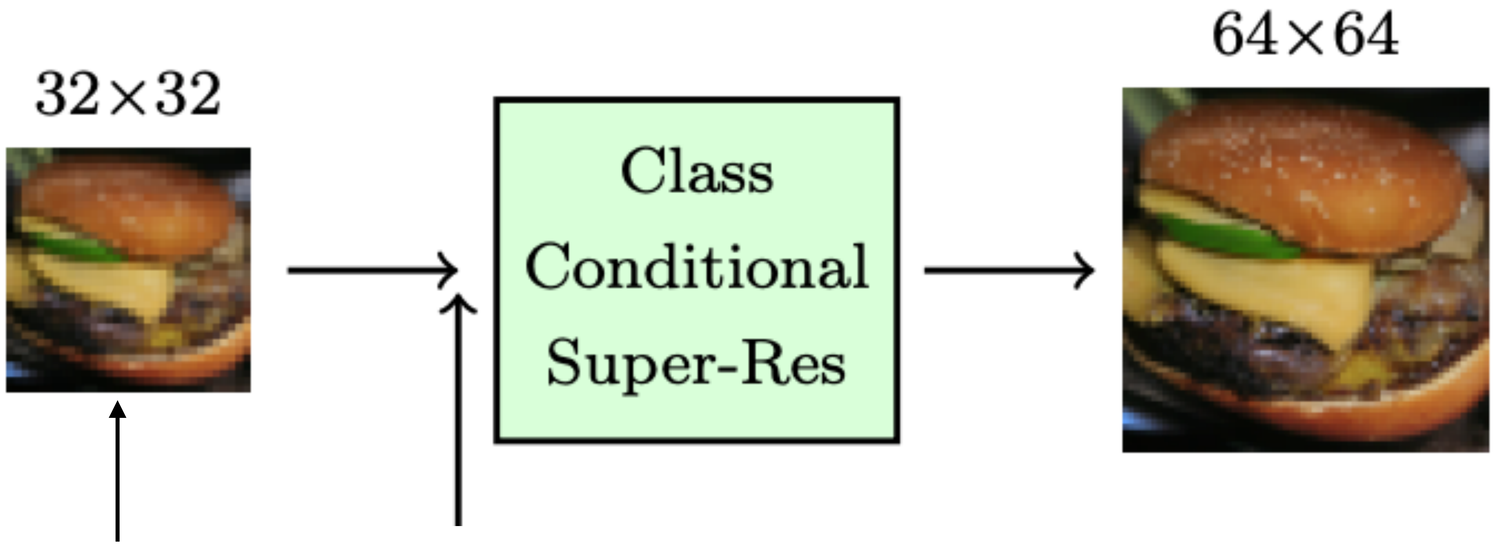


Image from: Cascaded Diffusion Models for High Fidelity Image Generation Ho et al., 2021

Training super-res models: Augmentation matters!



Augmentation Text caption

- Adding noise
- Blurring

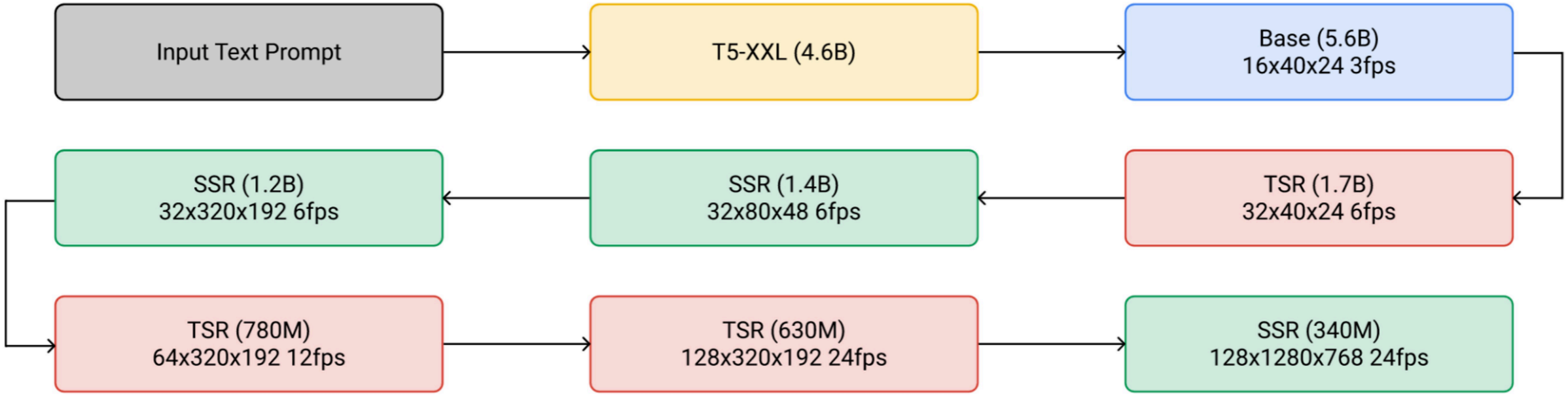
Trick borrowed from GAN papers

Image from: Cascaded Diffusion Models for High Fidelity Image Generation Ho et al., 2021

Imagen video

Imagen Video - Ho et al., 2022

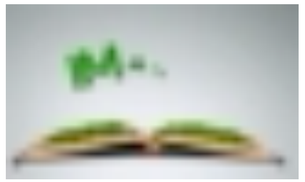
Imagen video



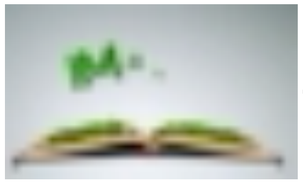
SSR = Spatial super resolution
TSR = Temporal super resolution
Cascade of 7 models (without text encoder)
Total parameters = 16.25B

Imagen video

Base 16x40x24 3fps
(5.6B)



TSR 32x40x24 6fps
(1.7B)



SSR 32x80x48 6fps
(1.4B)

SSR 32x320x192 6fps
(1.2B)

TSR 64x320x192 12fps
(780M)

TSR 128x320x192 24fps
(630M)

SSR 32x1280x768 6fps
(340M)

Imagen video

- Separable temporal and spatial convolutions
- Separable temporal and spatial attentions

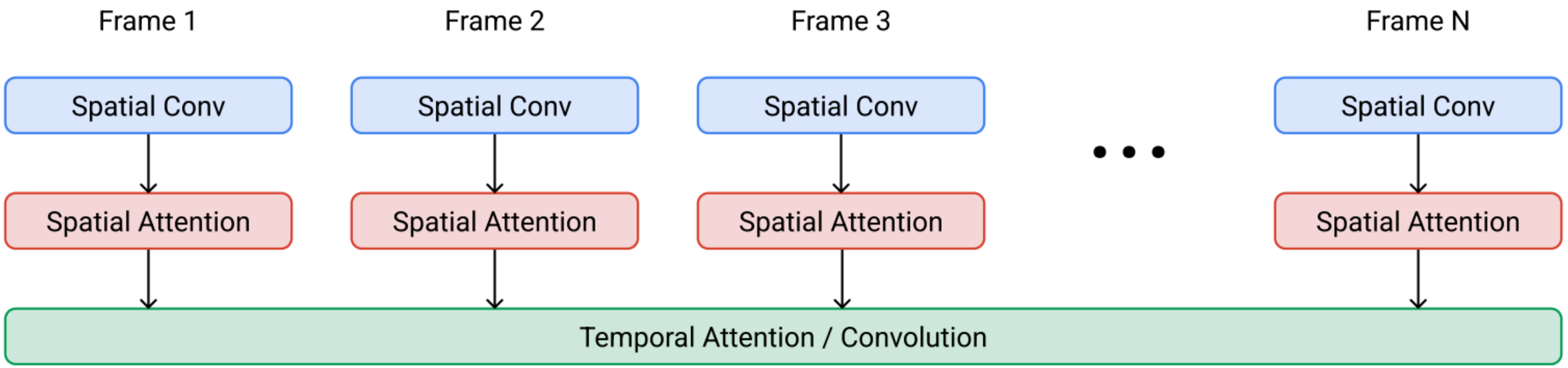


Imagen video

- Generates videos in different styles

A cat eating food out of a bowl, in style of Van Gogh.

A drone flythrough over a watercolor painting of a forest.

Drone flythrough of a pixel art of futuristic city.

Imagen video

- 3D structure

A 3D model of a 1800s victorian house. Studio lighting.

A 3D model of a car made out of sushi. Studio lighting.

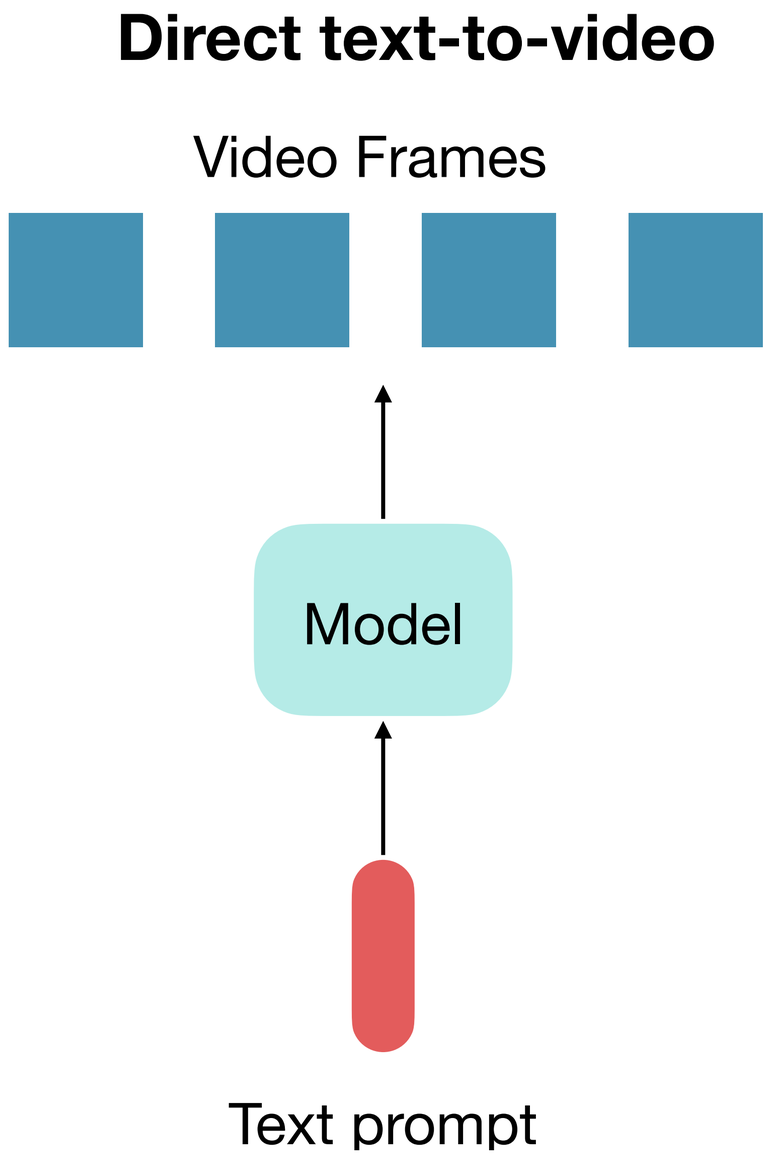
A 3D model of an elephant origami. Studio lighting.

Emu Video: Factorizing Text-to-Video Generation by Explicit Image Conditioning

Rohit Girdhar*^, Mannat Singh*^, Andrew Brown*, Quentin Duval*,
Samaneh Azadi*, Sai Saketh Rambhatla, Akbar Shah,
Xi Yin, Devi Parikh, Ishan Misra*

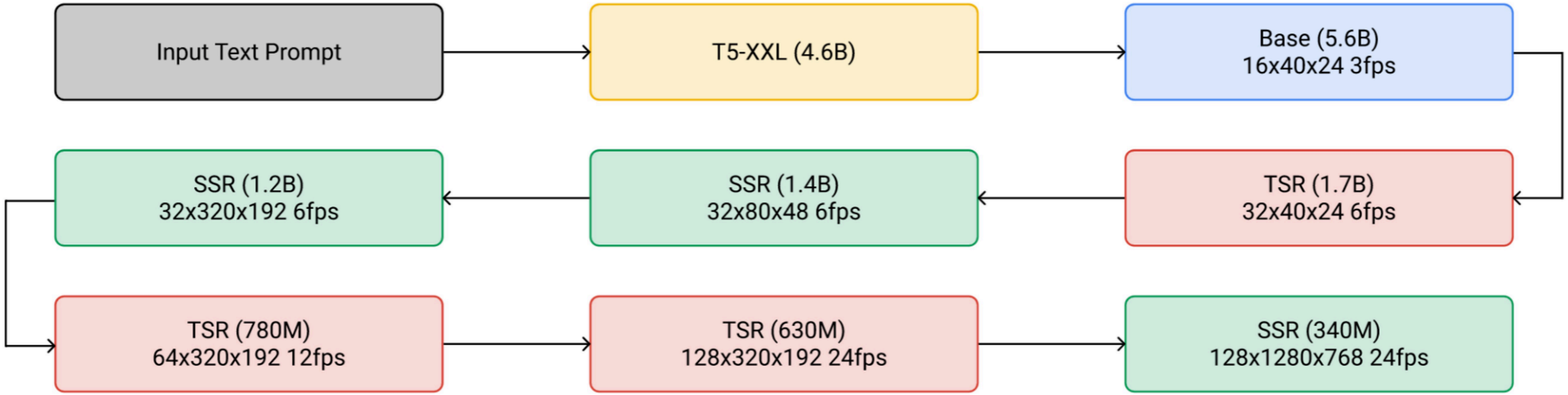
https://emu-video.metademolab.com/

Prior text-to-video diffusion approaches



- Directly generate “N” frames of a video conditioned on the text prompt
- Leverage a pretrained text-to-image model but lose its style when finetuning on videos
- May require deep cascades of models

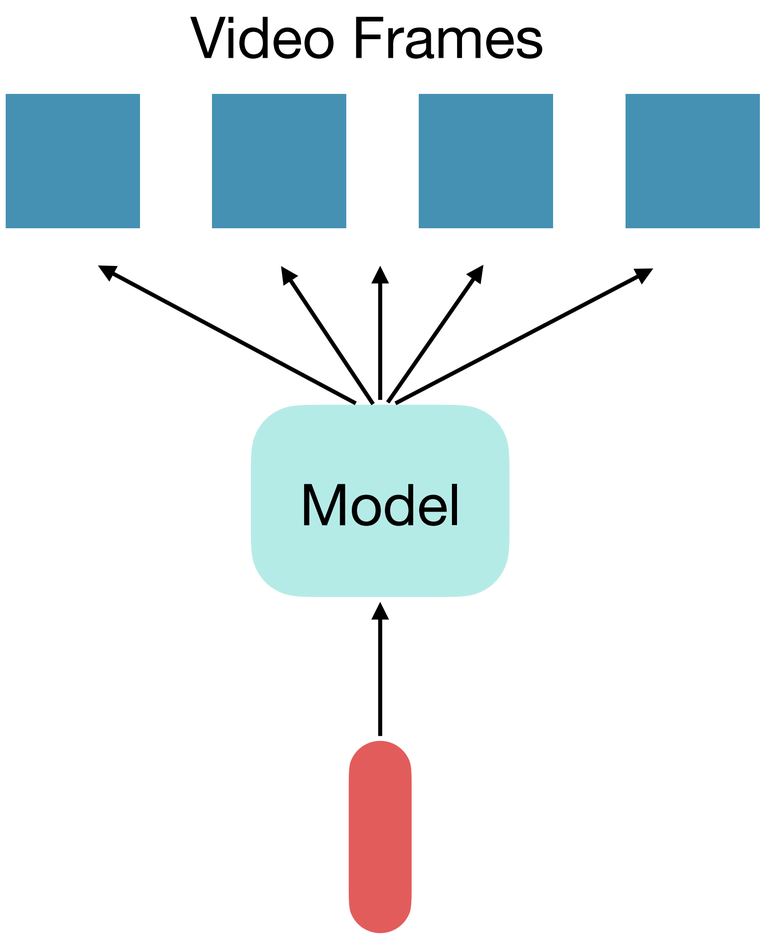
Imagen video



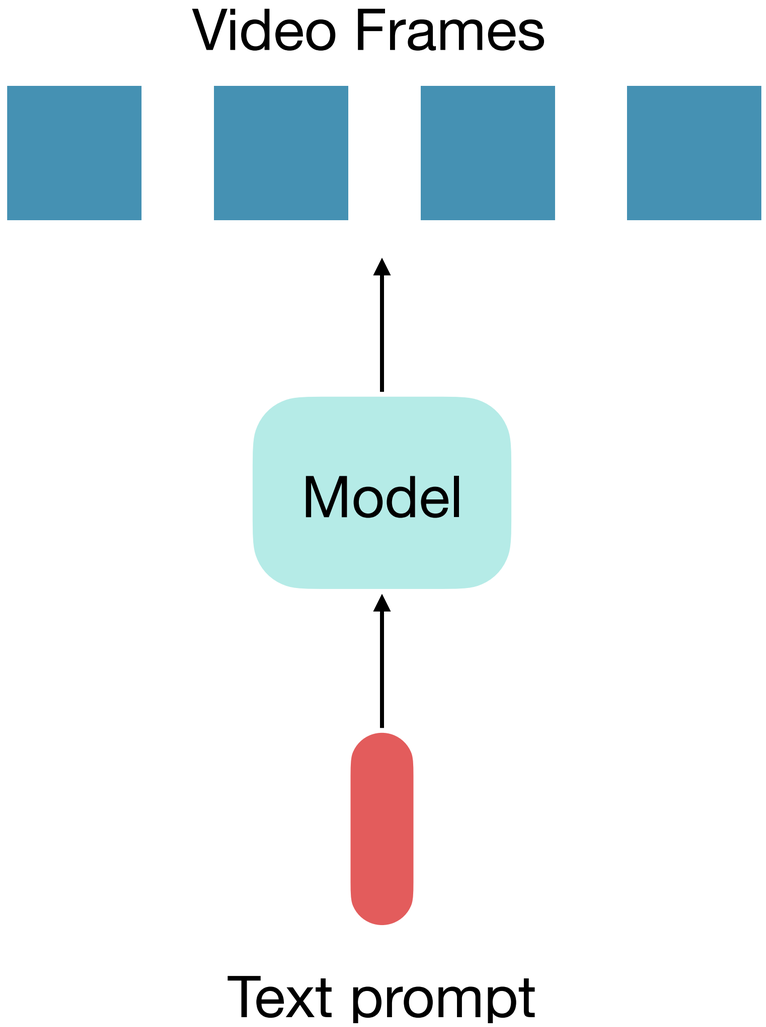
SSR = Spatial super resolution
TSR = Temporal super resolution
Cascade of 7 models (without text encoder)
Total parameters = 16.25B

Key insight: Factorized image-conditioned generation

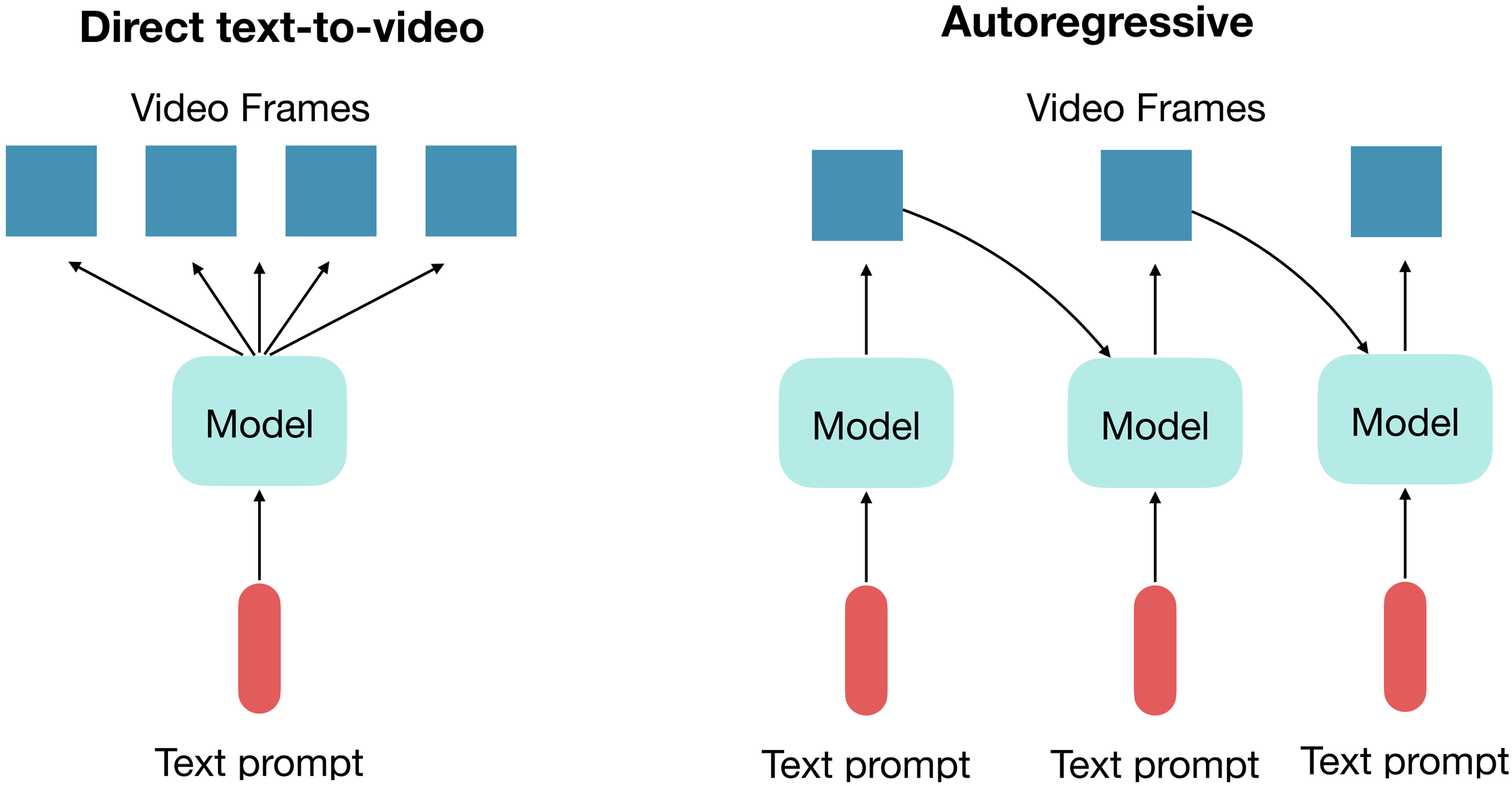
Direct text-to-video



Direct text-to-video

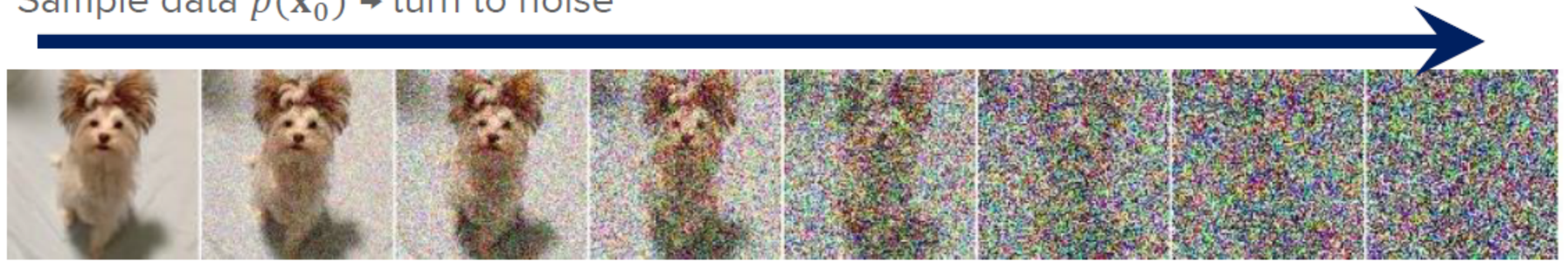


Why generate all frames at once?



Diffusion Model

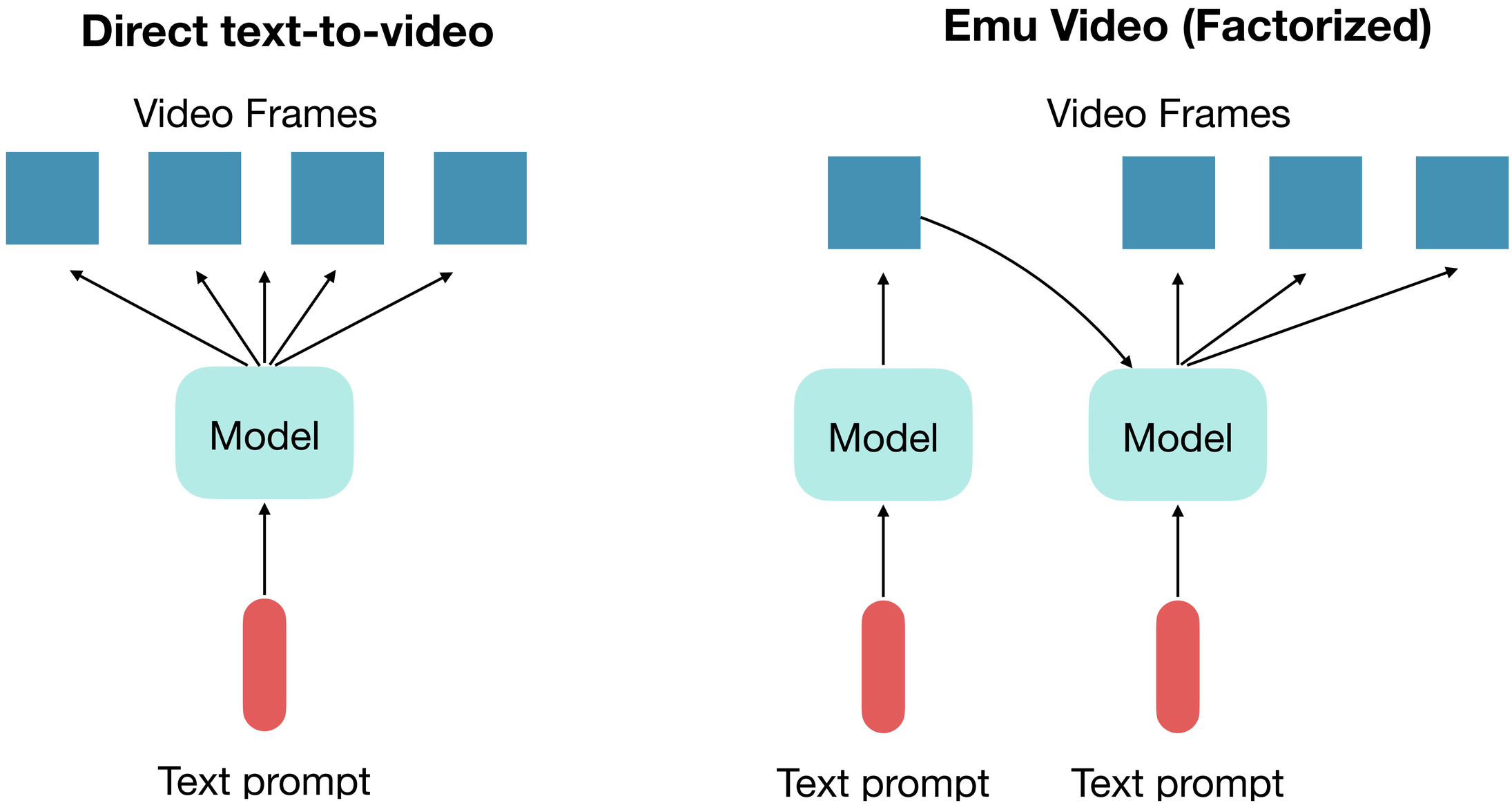
- **Forward / noising process**
  - Sample data $p(\mathbf{x}_0)$ → turn to noise



- **Reverse / denoising process**
  - Sample noise $p_T(\mathbf{x}_T)$ → turn into data

https://scholar.harvard.edu/binxuw/classes/machine-learning-scratch/materials/foundation-diffusion-generative-models

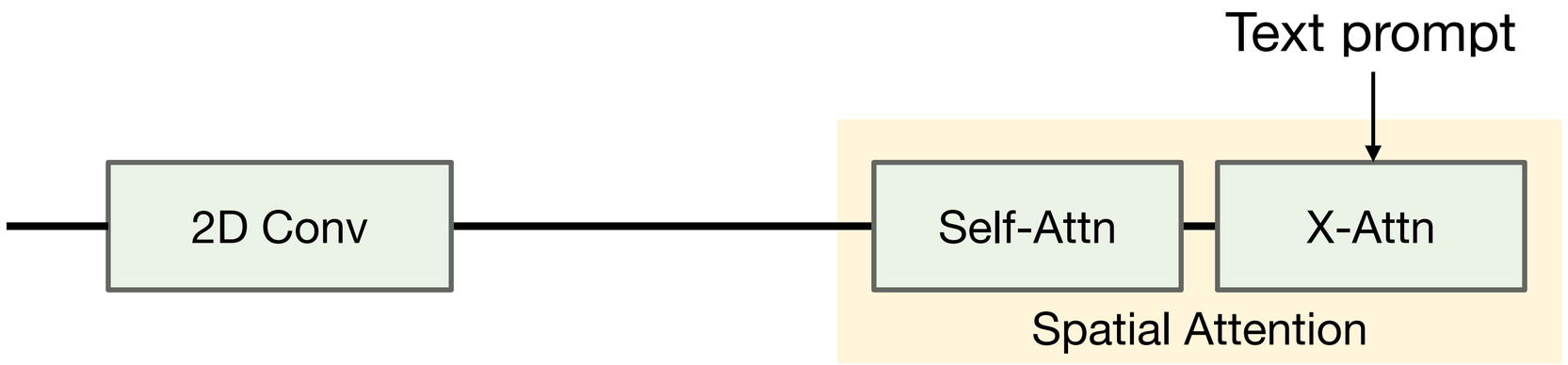
Key insight: Factorize by explicit image conditioning



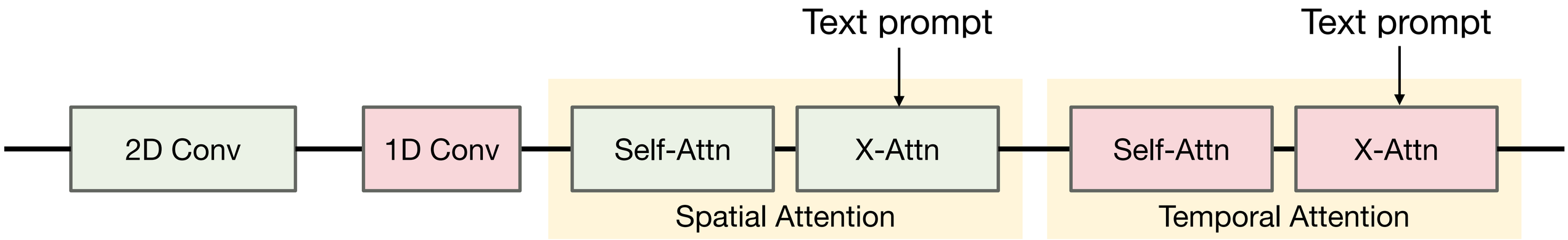
Emu Video

- Single text-to-video diffusion model (no need for deep cascades)
- Automatically supports image-to-video “animation”
- Retains style of pretrained text-to-image model trained on billions of images

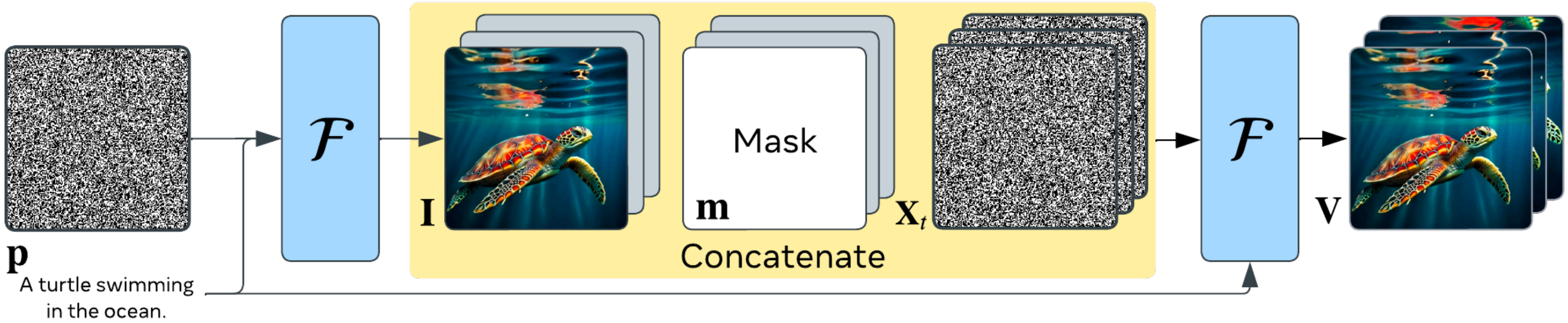
Adding temporal parameters to Diffusion models



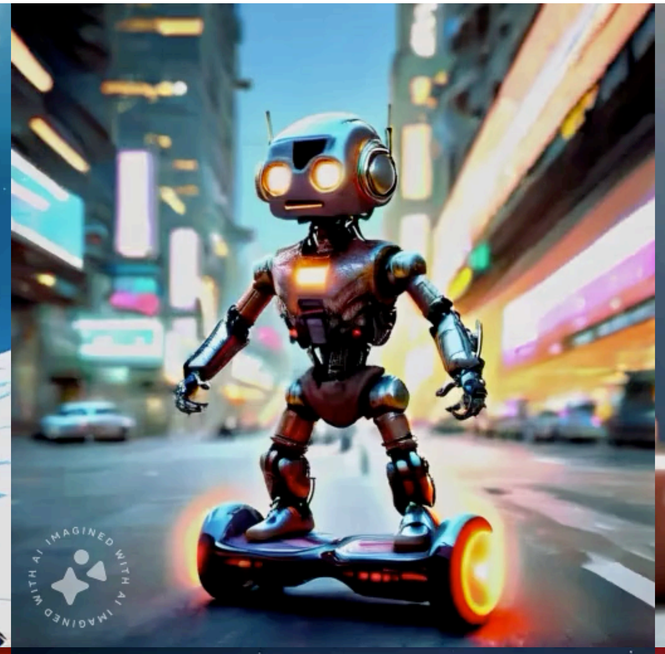
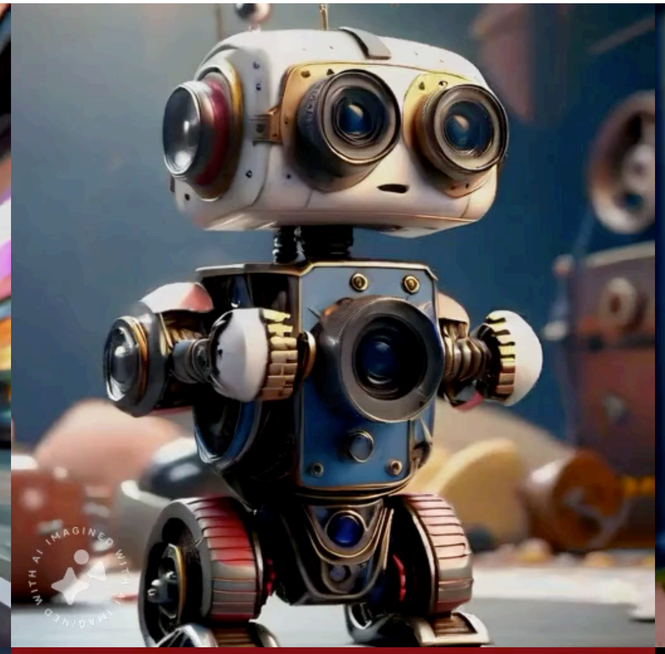
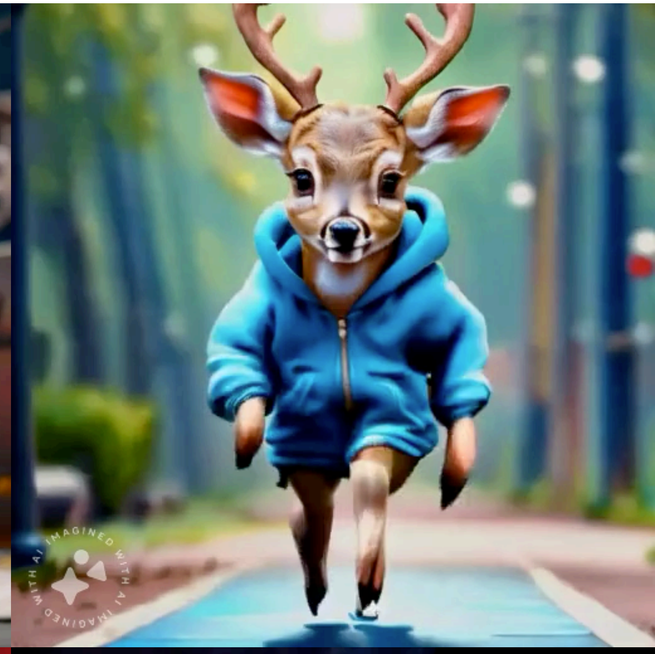
Adding temporal parameters to Diffusion models

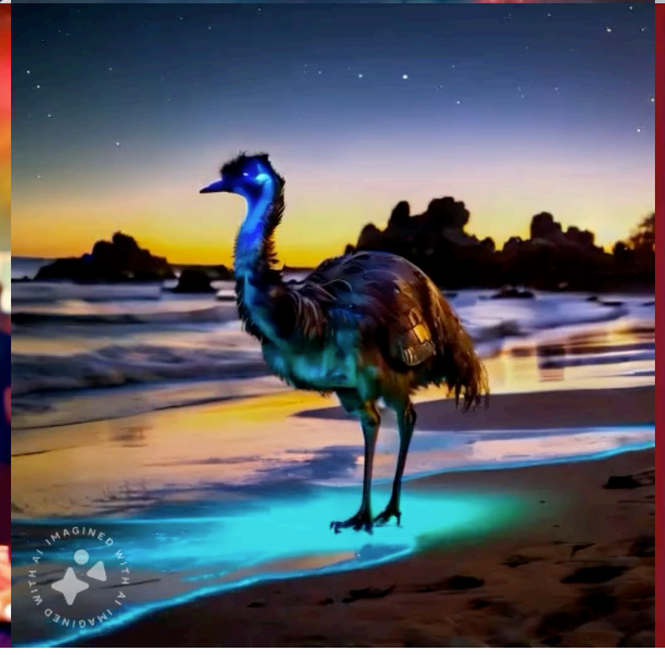


Emu Video



- 2.7B spatial parameters (frozen) + 1.7B temporal parameters (learned)
- Generates 16 frames at 4fps (4 seconds)
- Interpolation model converts 4fps to 16fps



Direct generation vs. Our Factorization

Direct

Factorized

Method	Quality	Faithfulness
Factorized	70.5	63.3

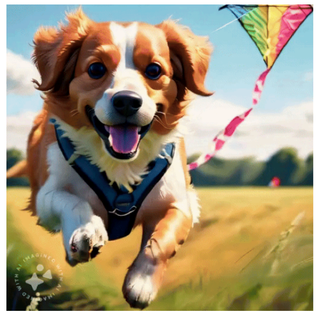
Image animation using the same model!

Movie Gen:
Foundation Model for Video Generation

GenAI, Meta

Google Imagen Video
Google Phenaki
Pika Labs
RunwayML Gen2

Meta Emu Video
Nov 2023



OpenAI SORA
Feb 2024

Gen3 Alpha
LumaLabs
Kling

Meta Movie Gen
Oct 2024

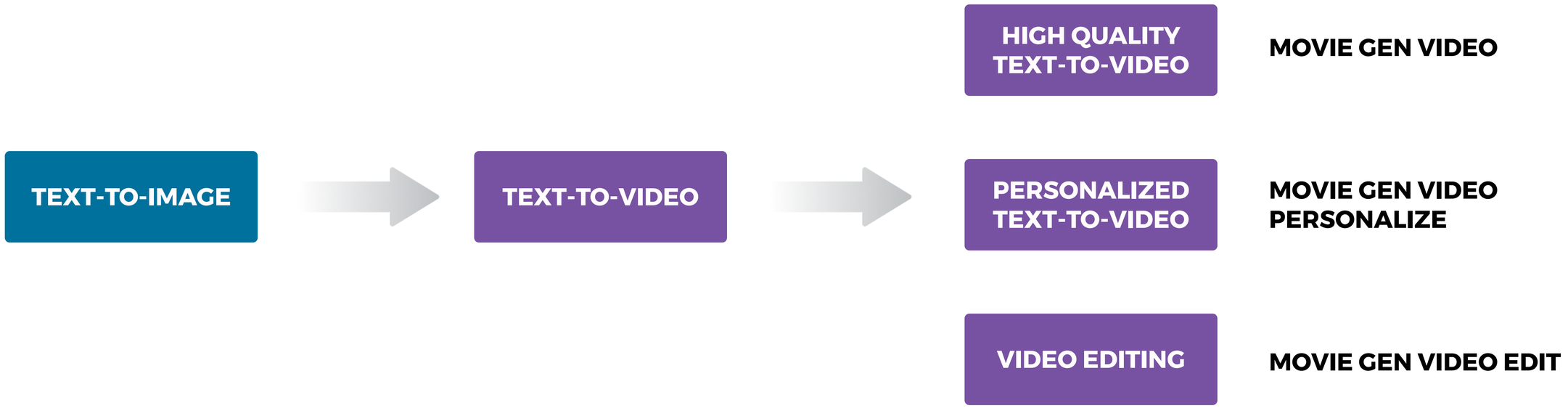
Movie Gen
Video Personalize

Movie Gen Video

Movie Gen
Video Edit

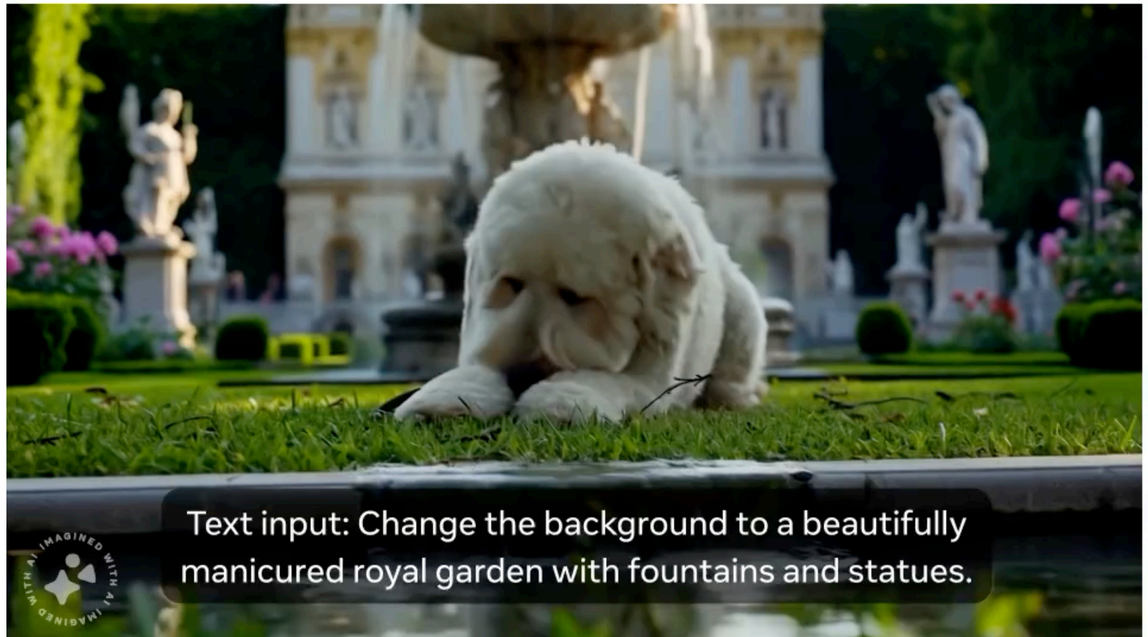
Image/Video sources from respective papers

Movie Gen Overview

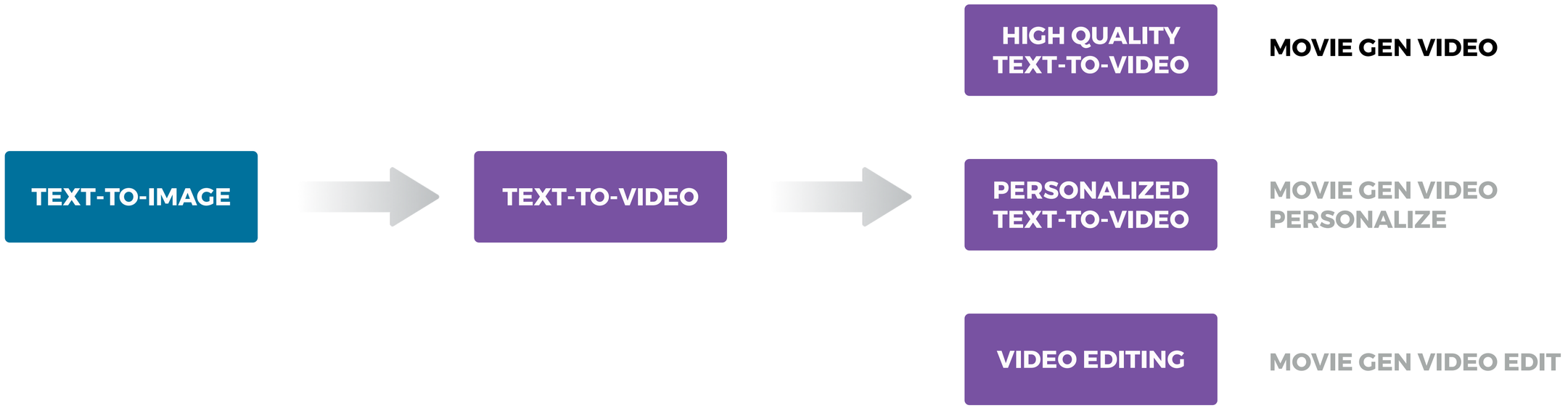


Movie Gen Video Personalize

Movie Gen Video Edit



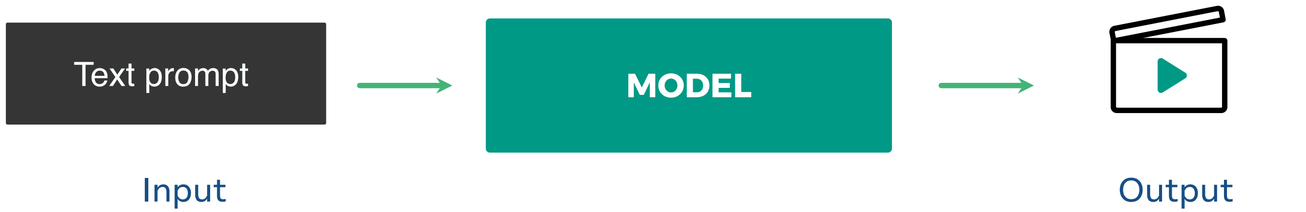
This talk: Movie Gen Video



Movie Gen Audio generates audio for a video
Detailed technical report (92 pages) released at https://arxiv.org/abs/2410.13720

Challenges in video generation

Low dimensional input to high dimensional output



Few words

TeraFLOPs per generation

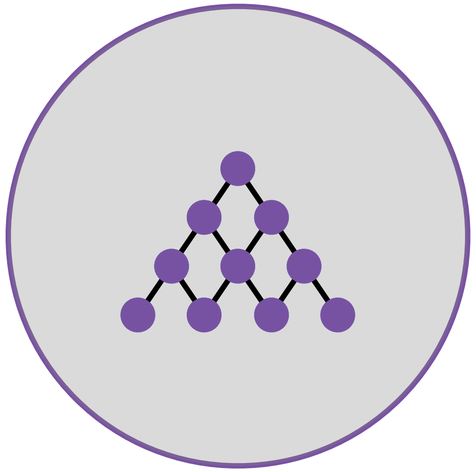
Millions of pixels

Visual consistency and quality

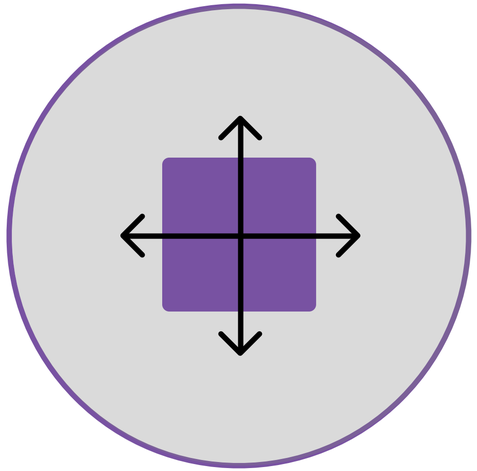
Computational Complexity

500M pixels for a 10 second HD video

Movie Gen: Key Improvements



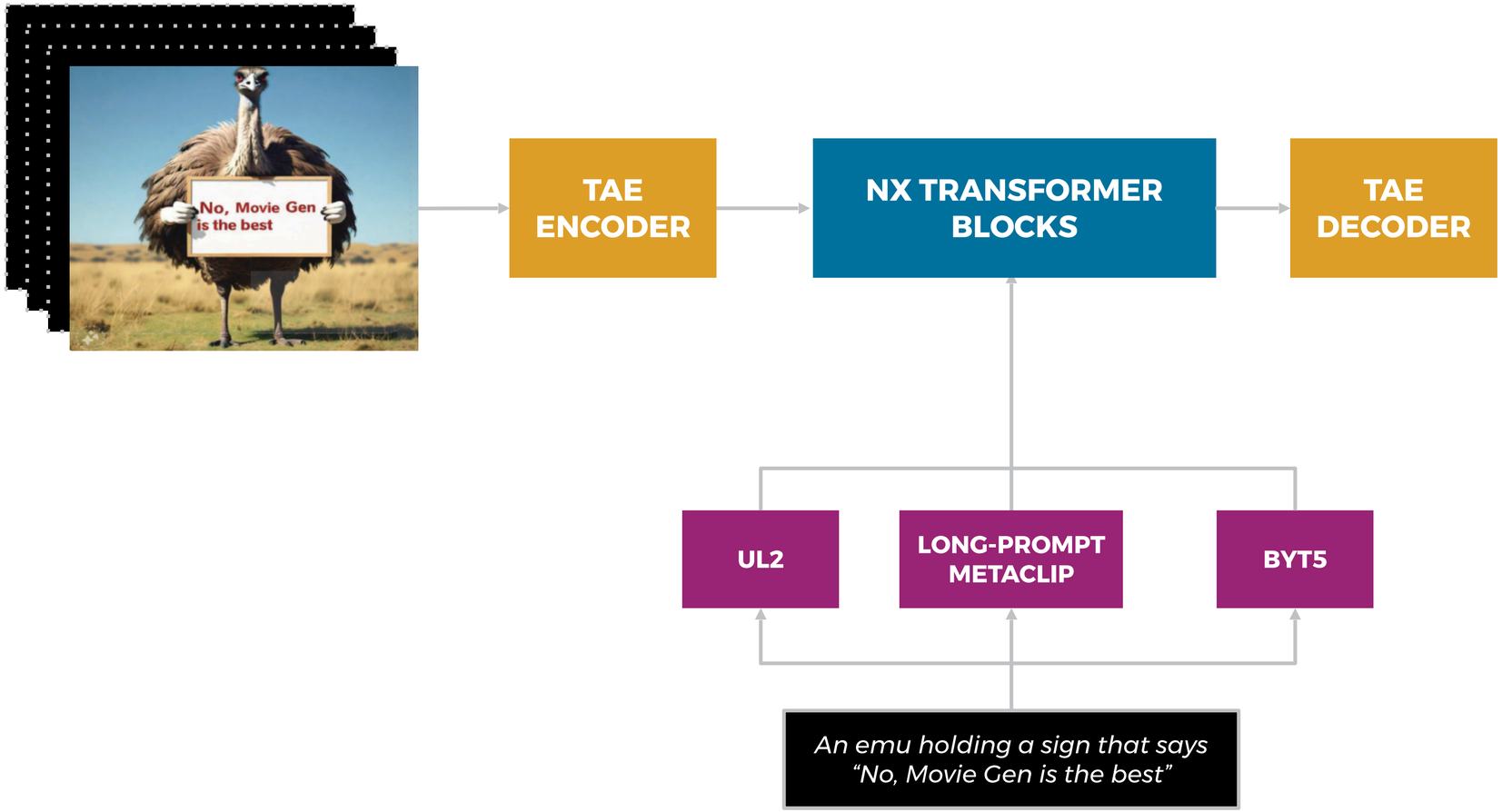
Simpler model architecture & training objective



Scaling up model, compute, data

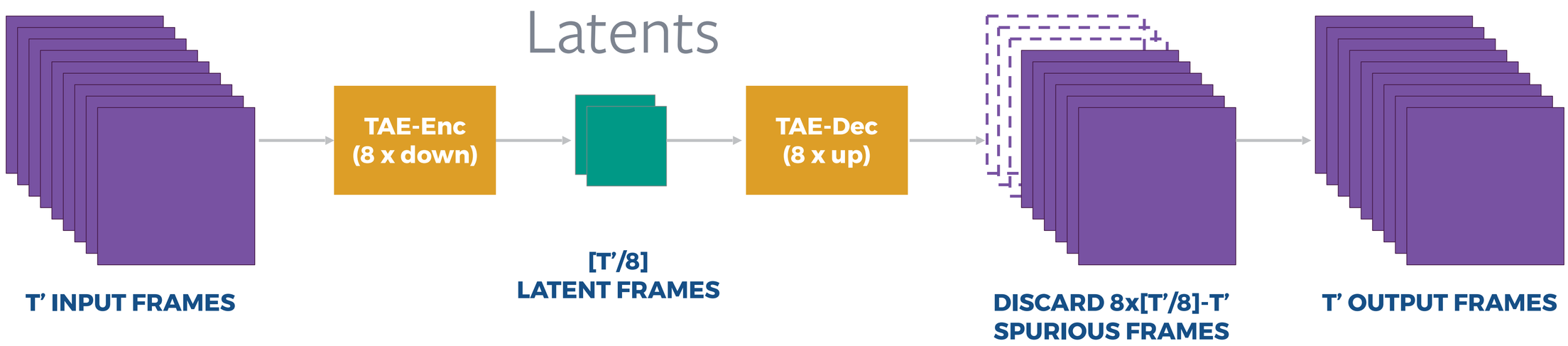
Improved training data quality

Simplified and Improved Architecture



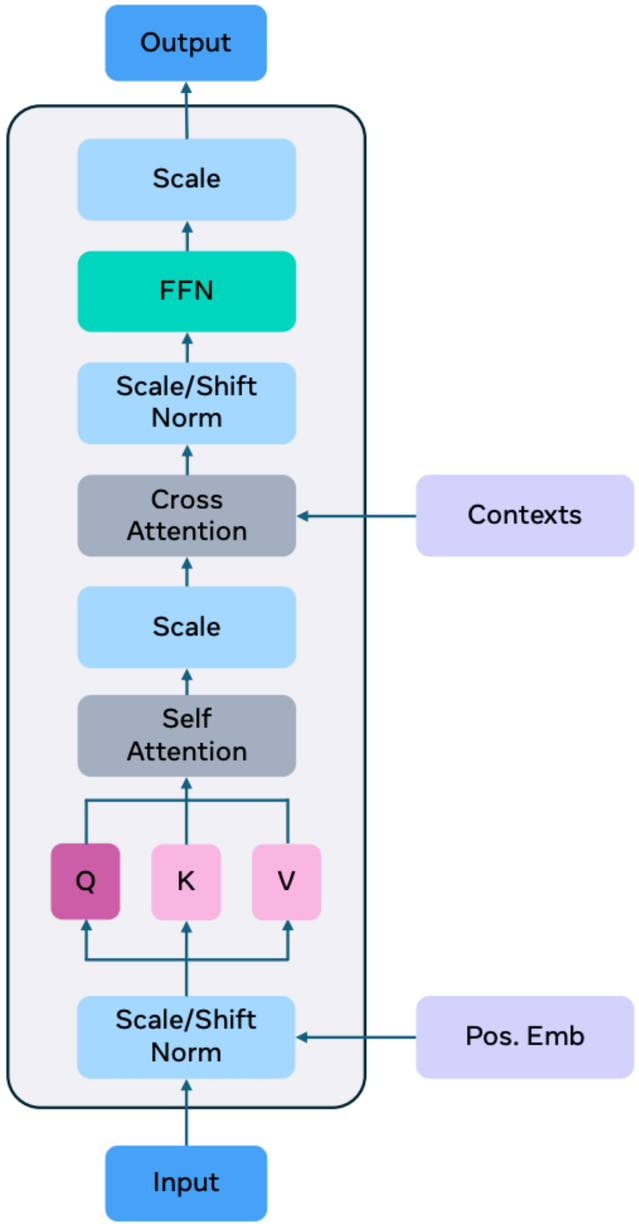
- Improved text encodings capture different aspects of the input text prompt
- Llama-style Transformer blocks make scaling easy

Improved Architecture: Compressed Latents



- Compress video by 8x across H, W, T
- 512x reduction in sequence length
- Compresses variable length sequences — videos or images

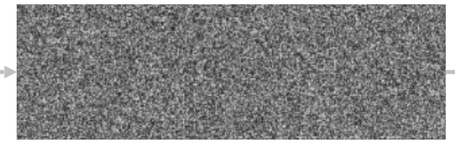
Improved Architecture: “Llama” style blocks



- Follow the Llama LLM block + DiT scale/shift
- Leads to stable training at scale
- All parameters trained on images and videos. No frozen parameters.

Improved Training Objective: Flow Matching

ORIGINAL VIDEO



GAUSSIAN NOISE

INTERPOLATE

NOISED VIDEO

NOISED VIDEO

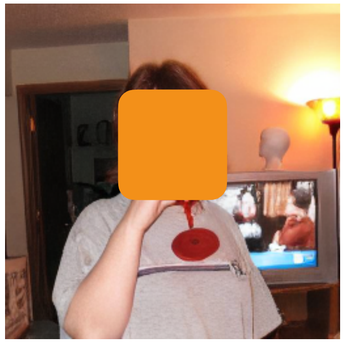
MODEL

Difference between
Original Video and Noised Video

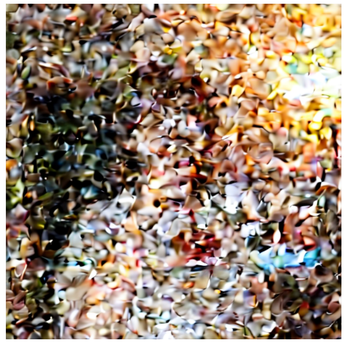
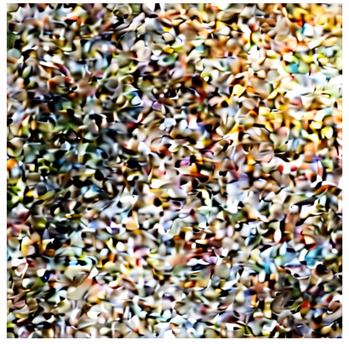
Captures the “direction” in which the noised video needs to be changed to make it the original video

Zero terminal signal-to-noise ratio (SNR) for Diffusion

256px



512px

Some structure still preserved at this step

Lin *et al. Common Diffusion Noise Schedules and Sample Steps are Flawed.* arXiv'23

Experiment from our prior work: Emu Video (2023)

W/o Zero SNR

With Zero SNR

Flow Matching vs. Diffusion

- Flow Matching is most similar to “v-prediction” in diffusion
- Flow Matching is “simpler” —> solves an ODE vs. SDE (diffusion)
- Flow Matching is naturally “zero terminal SNR”

Construct training sample

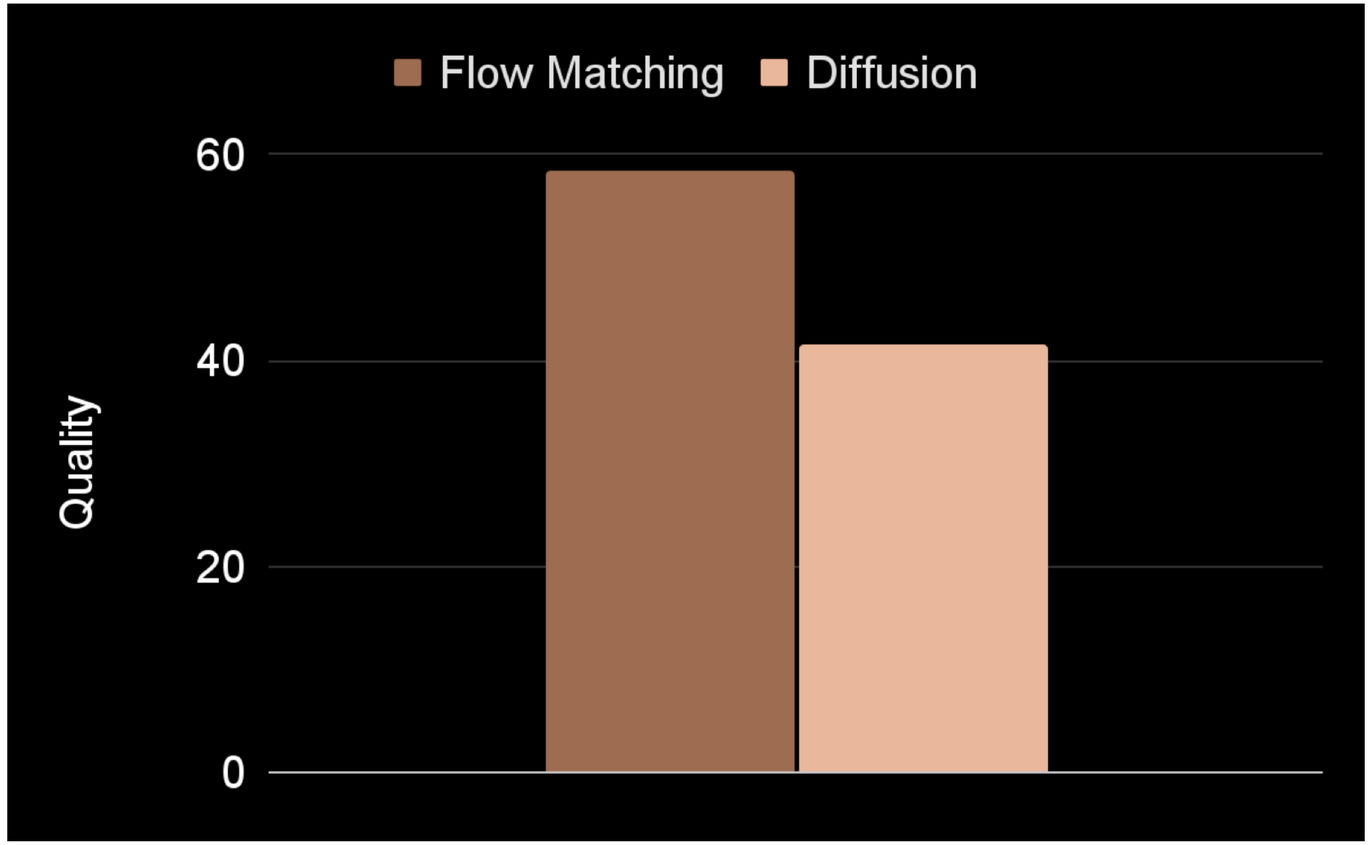
Noise Video

$$\mathbf{X}_t = t\ \mathbf{X}_1 + (1 - (1 - \sigma_{\text{min}})t)\ \mathbf{X}_0,$$

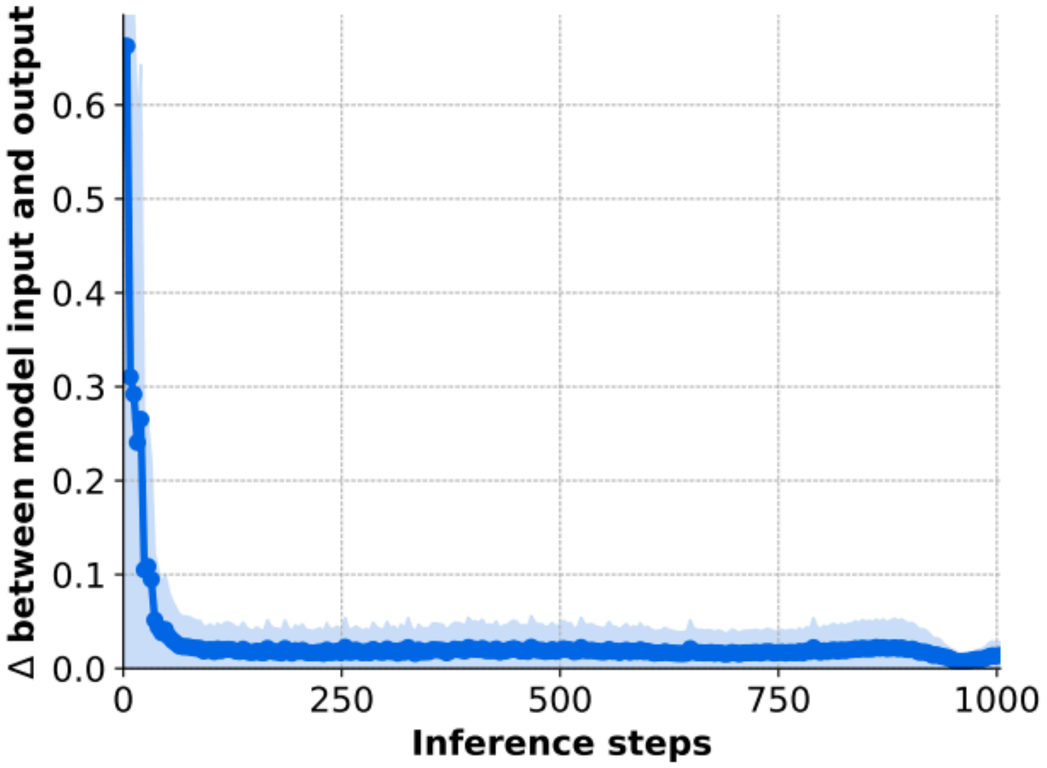
Ask the model to predict

$$\begin{aligned}\mathbf{V}_t &= \frac{d\mathbf{X}_t}{dt} \\ &= \mathbf{X}_1 - (1 - \sigma_{\text{min}})\mathbf{X}_0.\end{aligned}$$

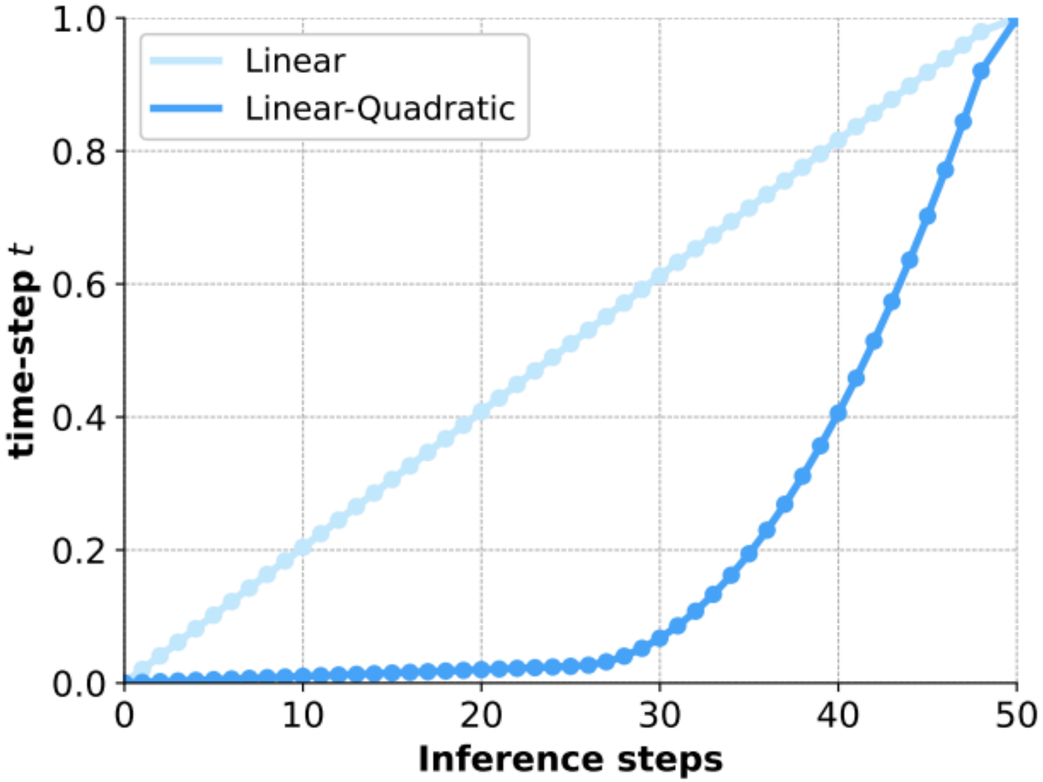
Flow Matching vs. Diffusion



Faster Inference



(a) Average change in model outputs between inference steps



(b) The linear-quadratic t-schedule

Scaling up

MODEL

10X LARGER

30B parameter model

Based on the Llama3 Transformer Architecture

TRAINING COMPUTE

10X LARGER

6144 GPUs

Multi-resolution training

Custom CUDA and GPU parallelism implementations

TRAINING DATA

10X LARGER

O(100M) video
O(1B) images

Data pipeline to curate high quality videos and images

VIDEO OUTPUT

4X LONGER

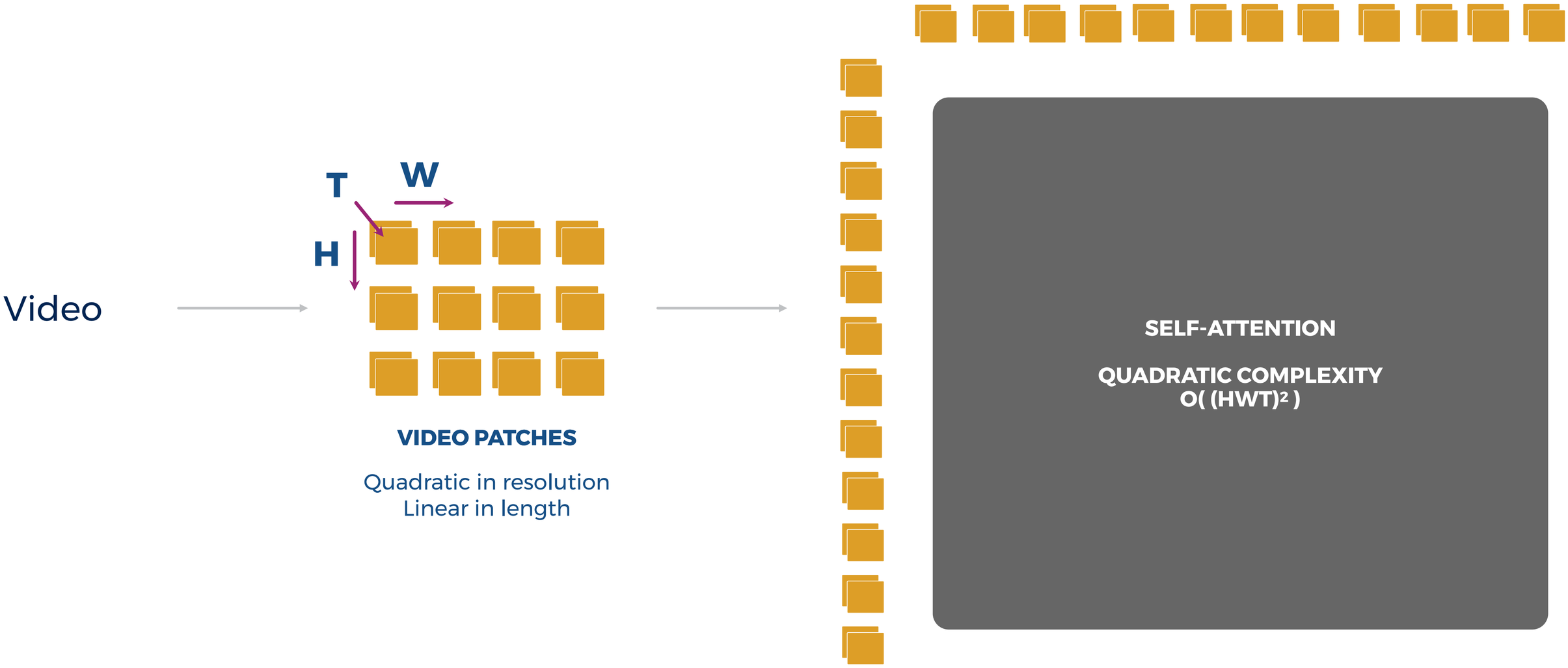
8X MORE PIXELS

Multiple aspect ratios

Higher frame rate

Scale wrt. Emu Video, 2023

Scaling up: Sequence Length



Multi-stage training

256px image training

256 tokens per image

256px image & video training

7K tokens per video

(16 seconds at 16 FPS, 256 x 256)

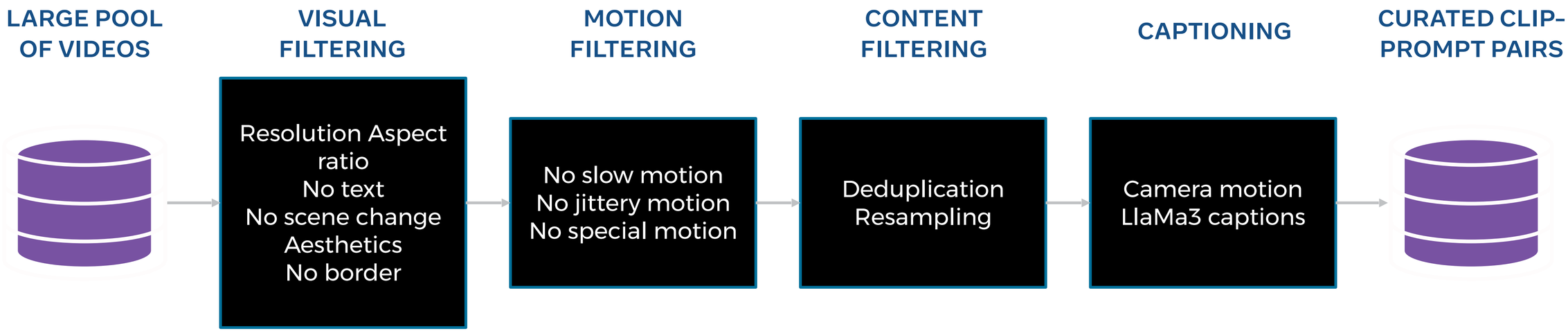
768px video training

73K tokens per video

(16 seconds at 16 FPS, 768 x 768)

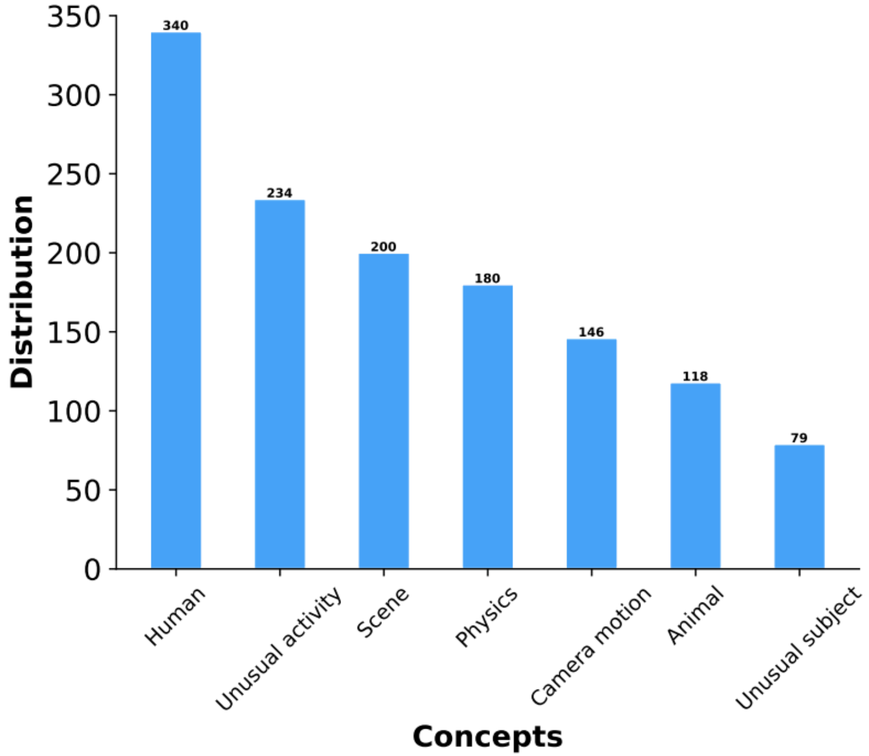
Video model produces 768px video. Separate spatial upsampler from 768px to 1080p

Training data improvements



- Training set consists of videos with different lengths, different fps
- O(100M) videos and O(1B) images
- Using Llama3 video captions improves text following ability

Evaluation Benchmark: Movie Gen Bench



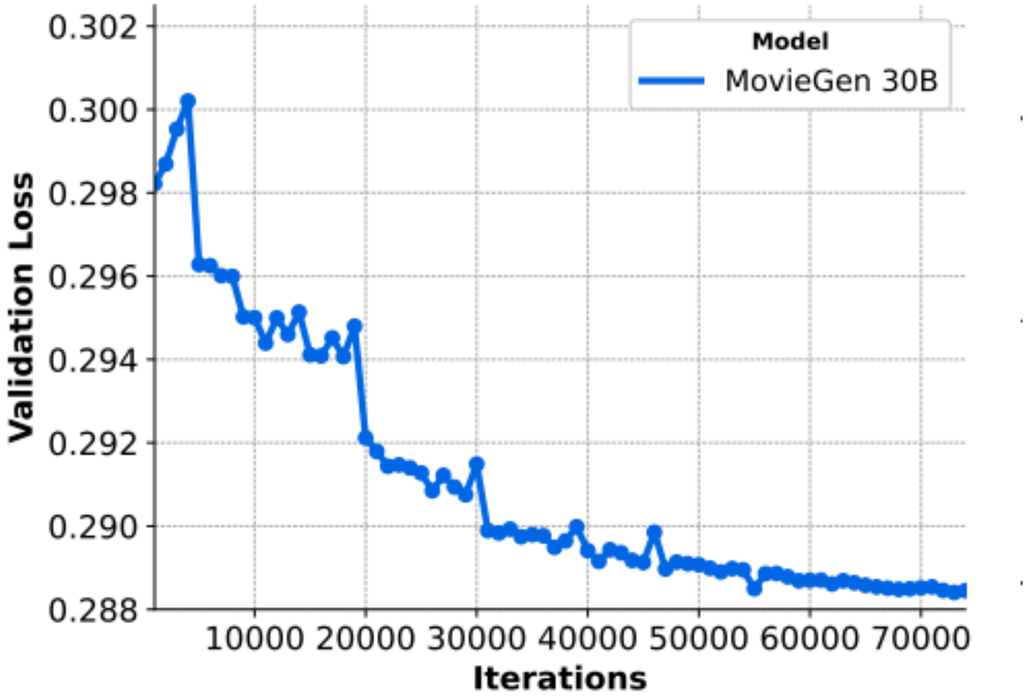
(a) Prompt concept distribution

(b) Nouns in evaluation set

(c) Verbs in evaluation set

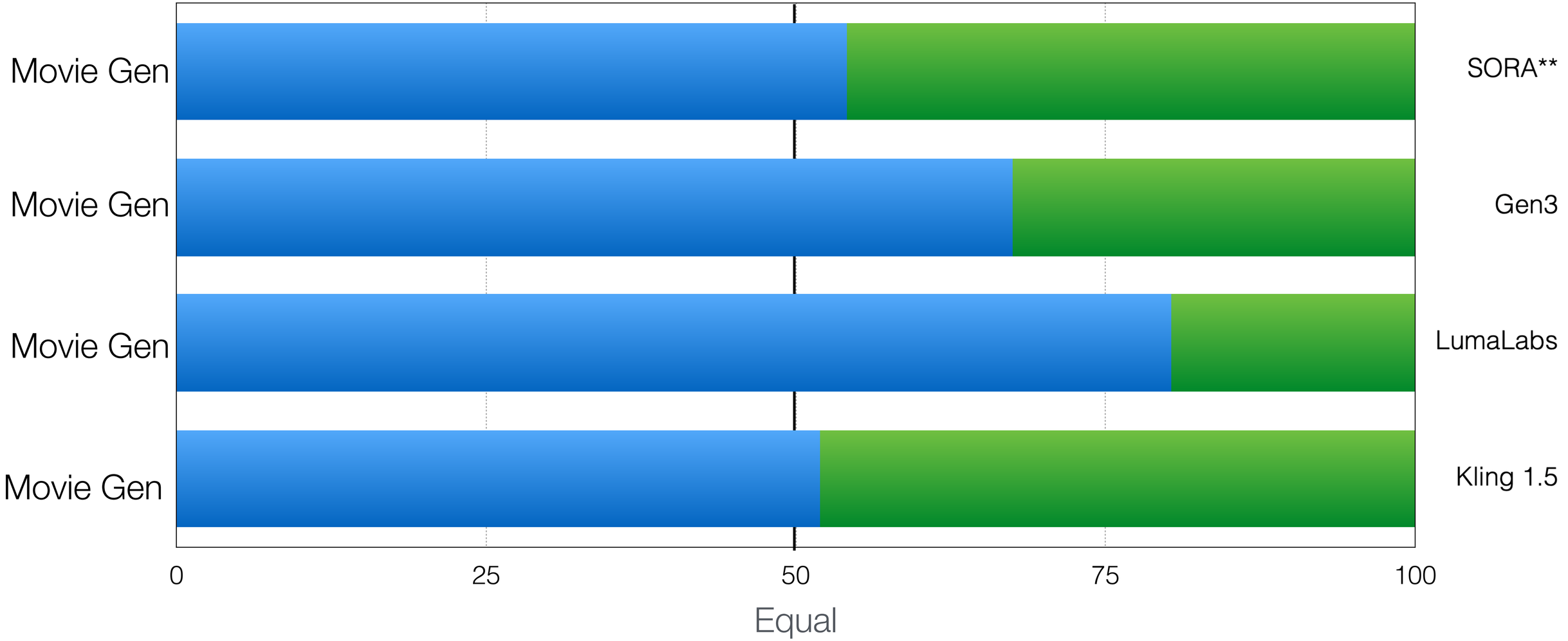
- 3x larger benchmark than prior work
- Diverse coverage of concepts
- Released on Github: 1000 videos. Not cherry picked
- https://github.com/facebookresearch/MovieGenBench

Validation Loss correlates with Human Judgment

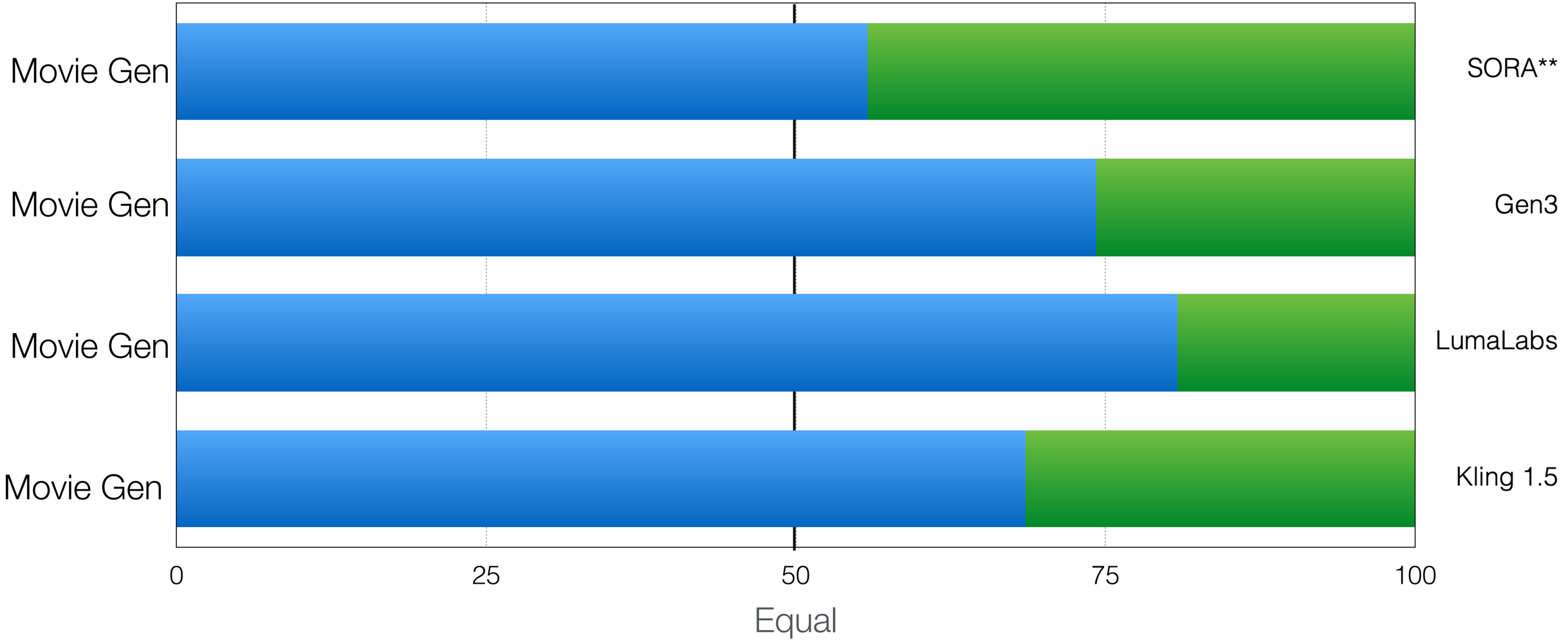


Model A iterations	Model B iterations	Model A net win rate *vs.* model B				
		Overall Quality	Frame Consistency	Motion Completeness	Motion Naturalness	Text-alignment
15.6k	10.2k	16.0	9.0	4.67	7.66	5.67
19.2k	15.6k	13.66	-0.67	5.34	5.66	3.34
37k	26.4k	0.39	-4.3	0.78	-1.95	3.13
59k	54k	6.33	-1.33	8.34	3.0	8.0

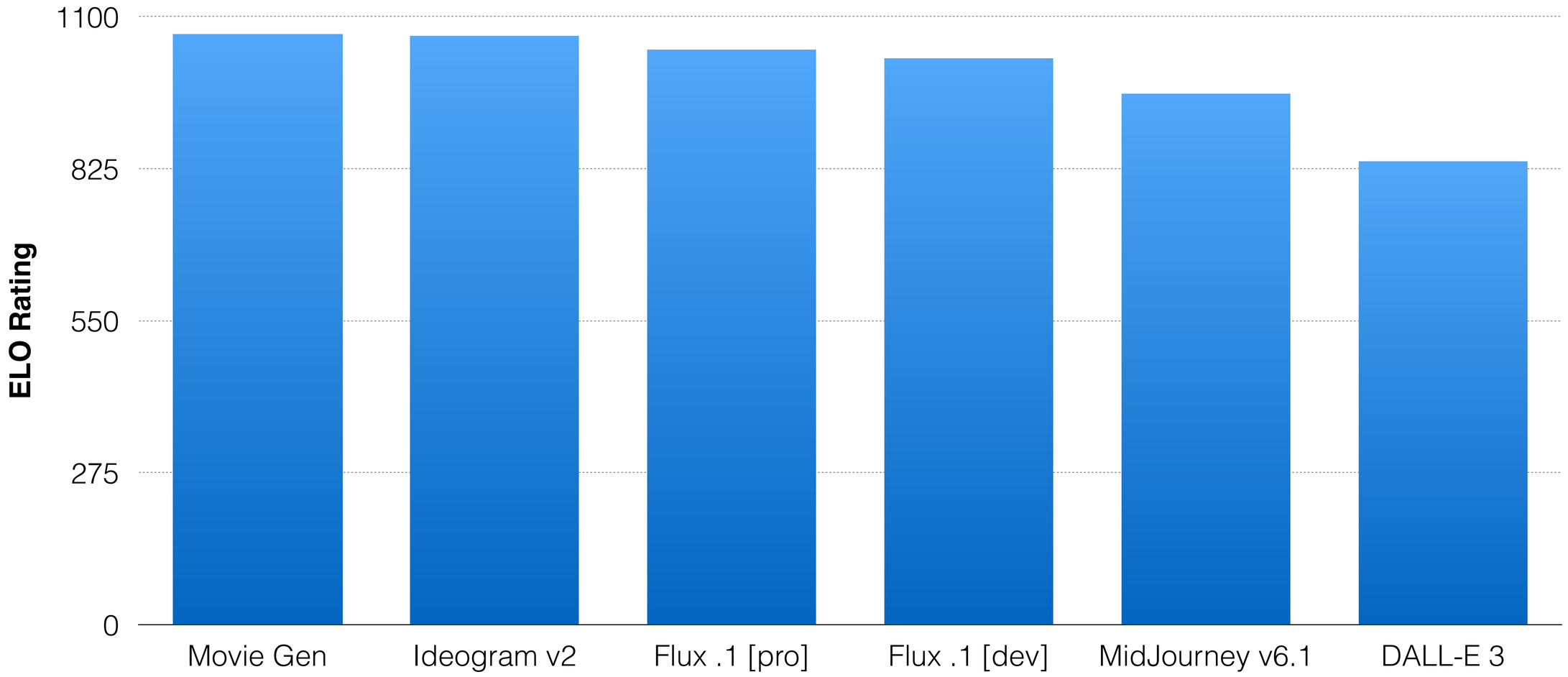
Evaluation Results: Overall Quality



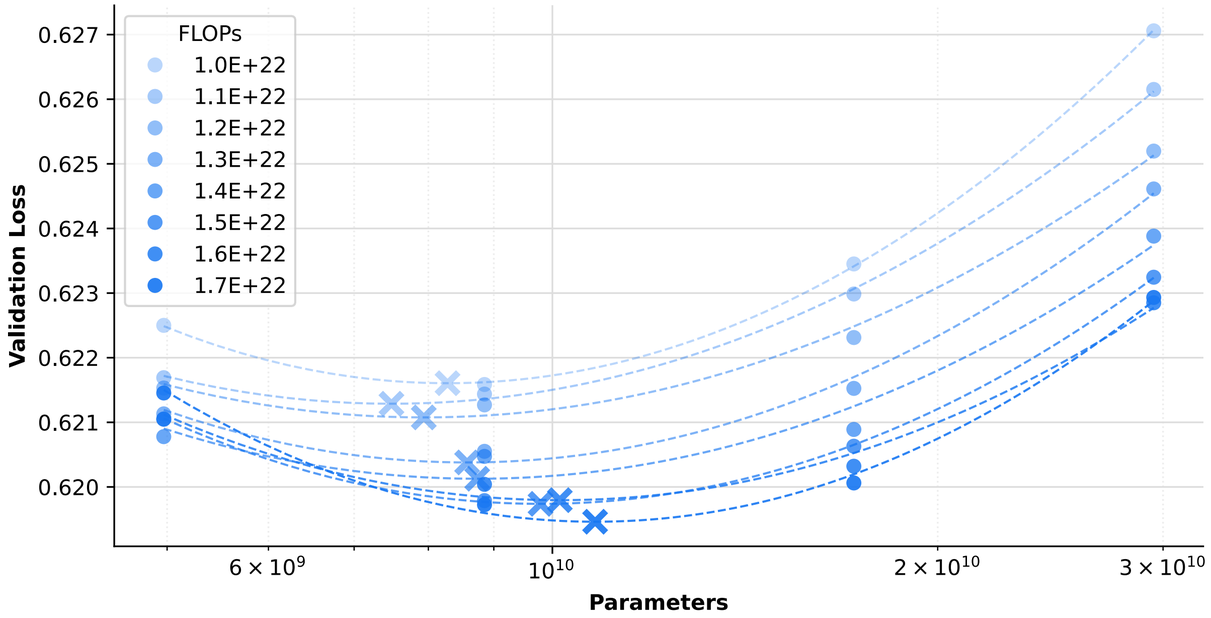
Evaluation Results: Realness

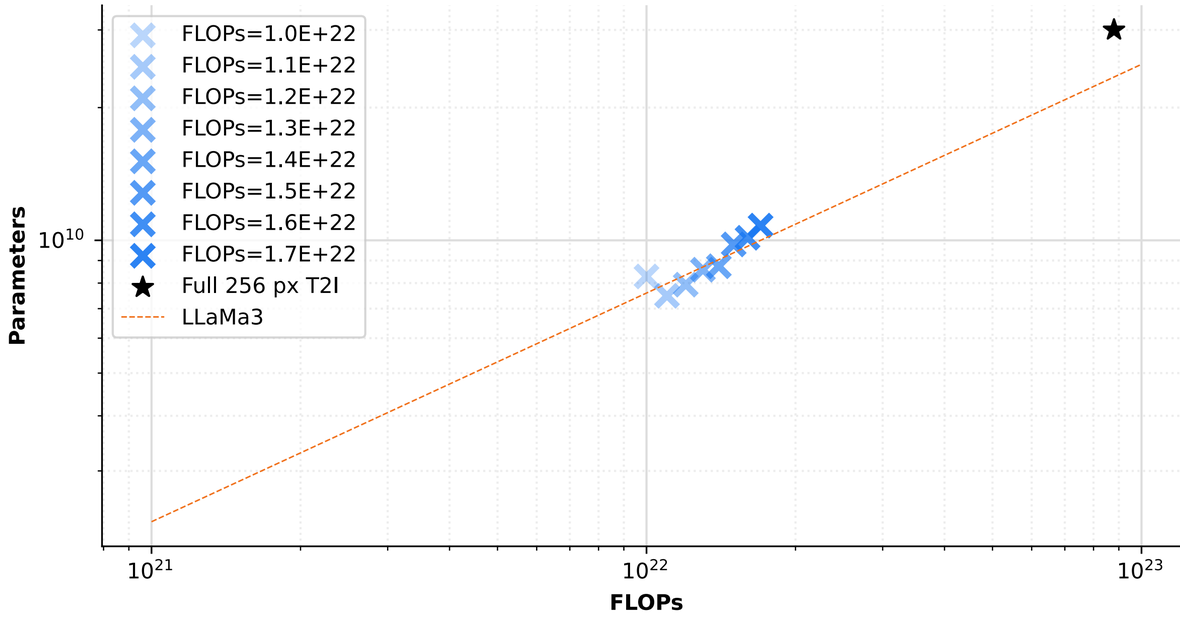


Evaluation Results: Text to image generation



Scaling Laws





- Performance does not seem saturated
- Scaling Laws for video generation closely follow LLMs (Llama3 in this case)

Thanks!